Demo Software User Manual

C# Versioin

V1.0

Table of Contents

1. Introduction 1
2. System Requirements 1
3. Installa and Uninstal 1
4. Quick Start 1
    4.1. Open the Software 1
    4.2. Switch Language 1
    4.3. Connect to a Device 2
    4.4. Inventory 3
5. Interface Description 5
    5.1. Interface Zones 5
    5.2. Menu Description 6
6. Software Functions 8
    6.1. Connection Management 8
        6.1.1. Connecting to a Device 8
        6.1.2. Disconnect the Device 10
        6.1.3. Device Restart 11
        6.1.4. Device Restore 11
        6.1.5. Command Debugging 11
        6.1.6. Device Upgrade 12
    6.2. Iventory 14
        6.2.1. Iventory Parameters 15
        6.2.2. Column Selection 17
        6.2.3. Clear 18
        6.2.4. Iventory 18
        6.2.5. Advanced Inventory 19
        6.2.6. Stop 21
        6.2.7. UHF Tag Operations 21
    6.3. Device Management 28
        6.3.1. Communication Settings 28
        6.3.2. Network Settings 30
        6.3.3. Wireless Settings 32

6.3.4. Buzzer & Indicator Light 35
6.3.5. GPIO Settings 37
6.3.6. Other Settings 39
6.4. RFID Settings 41
6.4.1. Antenna Settings 41
6.4.2. RFID Settings 42
6.4.3. UHF Settings 44
6.4.4. EAS Settings 48
6.4.5. Output Settings 49
6.5. System Settings 53
6.5.1. System Settings 53
6.5.2. Log 54
7. Common Issues & Solutions 55
7.1. Unable to Connect to Device via Serial Port 55
7.2. Unable to Connect to Device via Network Port 55
7.3. Unable to Find Device When Searching Network Port 55
8. Security and Privacy 55

1. Introduction

This software is a general reader management tool developed by our company, designed to help users efficiently test device functions and performance, and conveniently configure various parameters.

2. System Requirements

- Operating System: Windows 10/11
- Processor: Intel Core i3 or equivalent performance
- Memory: 8GB RAM (16GB recommended)
- Storage Space: At least 2GB of available space
- Other Dependencies: .NET Framework 4.0

3. Installa and Uninstal

This software is a standalone executable file; users do not need to install it.

4. Quick Start

4.1. Open the Software

Locate DeviceCenter , and double-click to start the software.

4.2. Switch Language

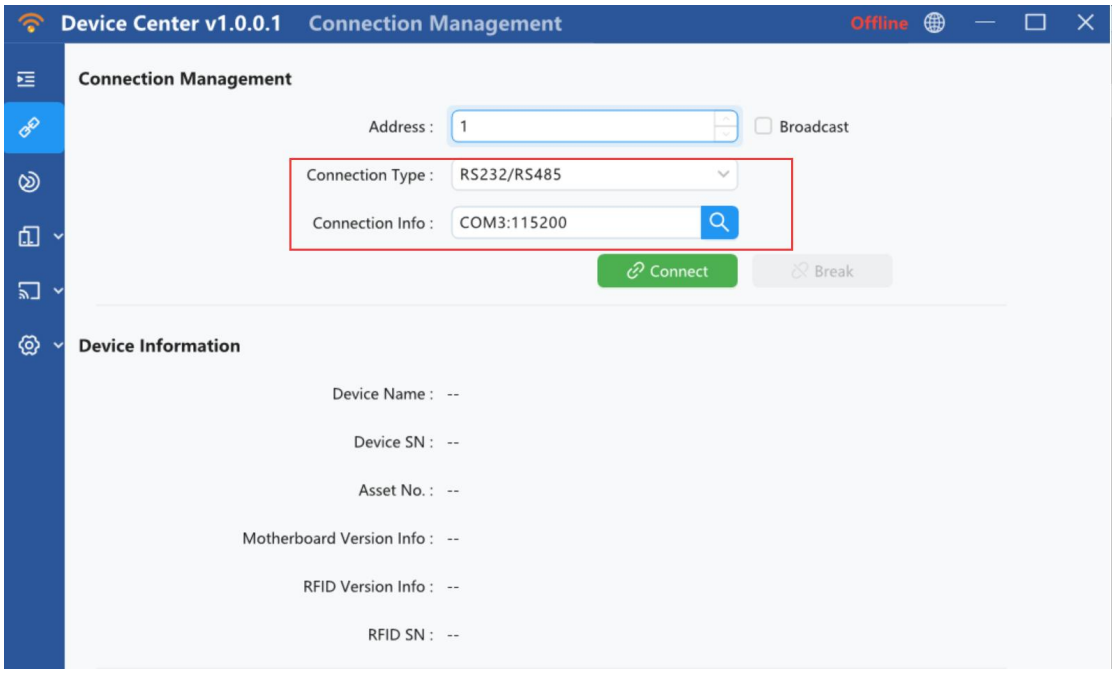
Click the language button in the upper right corner of the software interface to pop up the launguage selection menu， as shown below .

Select the language you need to switch to in the menu, and click to complete the switch. The software currently supports Simplified Chinese, Traditional Chinese, and English.

4.3. Connect to a Device

1、Ensure the device is properly powered on and has established a stable connection to the PC via RS232, RS485, Network Port, or USB.

2、In the interface shown below, select the appropriate connection method and modify the connection information according to the following formats:



3、RS232/RS485: Adjust the format to [Serial Port: Baud Rate], e.g., COM1:115200

4、TCP: Adjust the format to [IP:Port], e.g., 192.168.1.201:8090

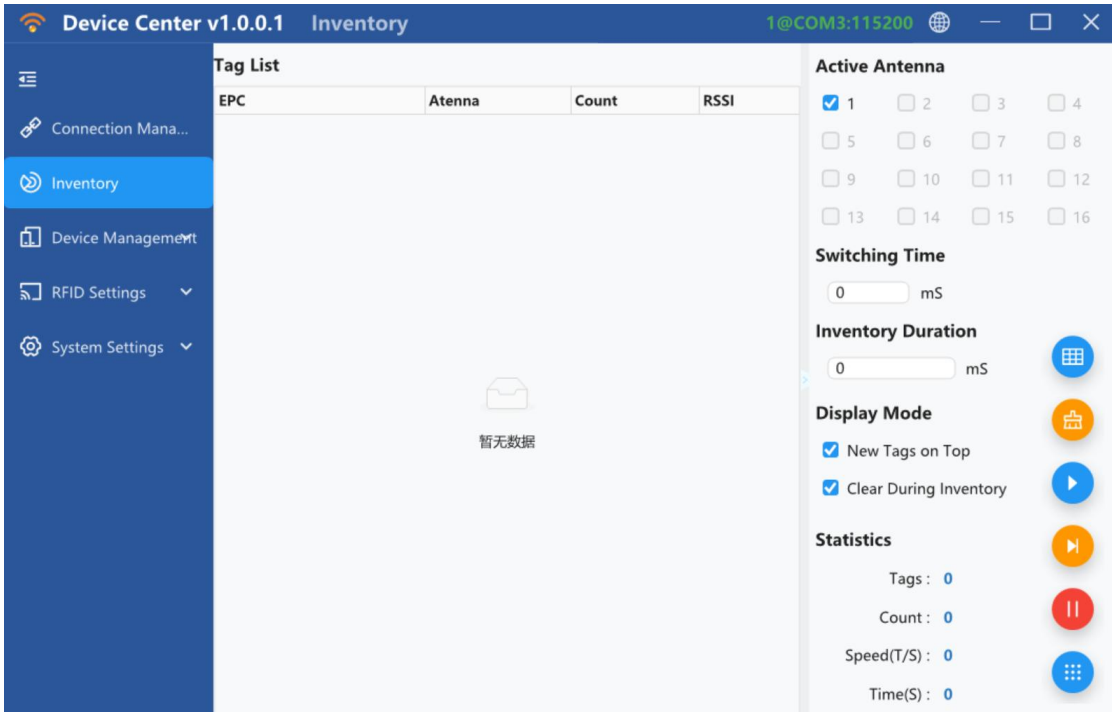
5、Click the "Connect" button

to attempt to establish a connection. After a successful connection, the interface will appear as follows:

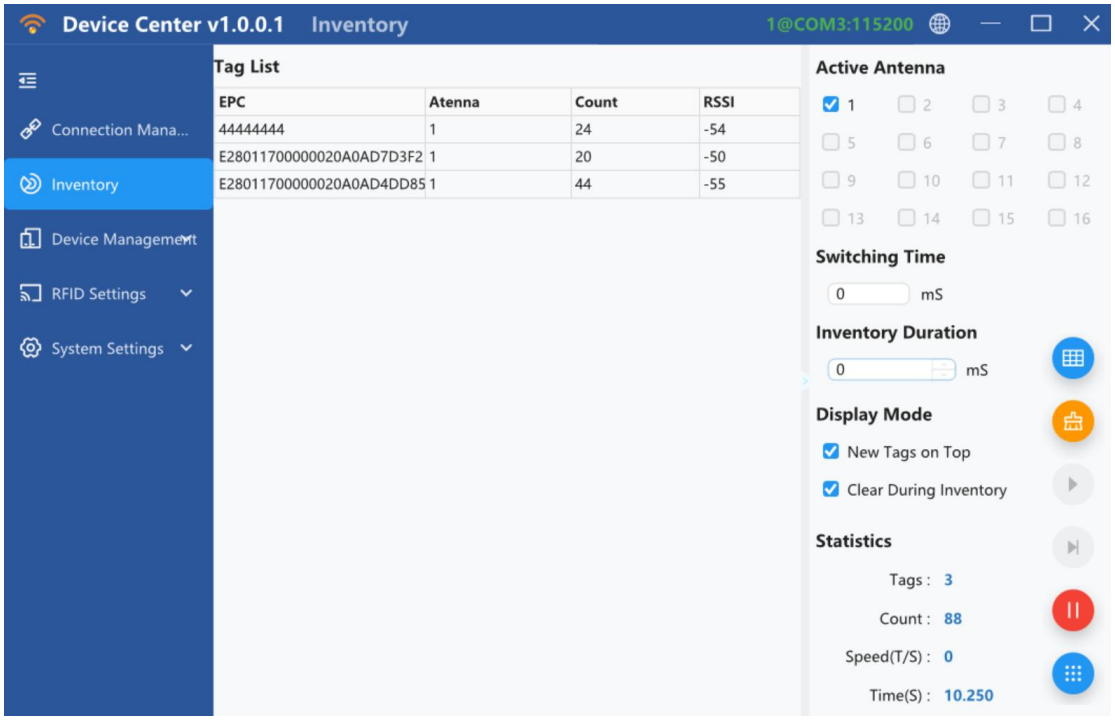
4.4. Inventory

1、After a successful connection, click the "Inventory" menu

in the menu bar to switch to the Inventory page. The specific operations are as follows:



2、Click the "Inventory" button to start the inventory function.Detailed steps are shown in the figure below.



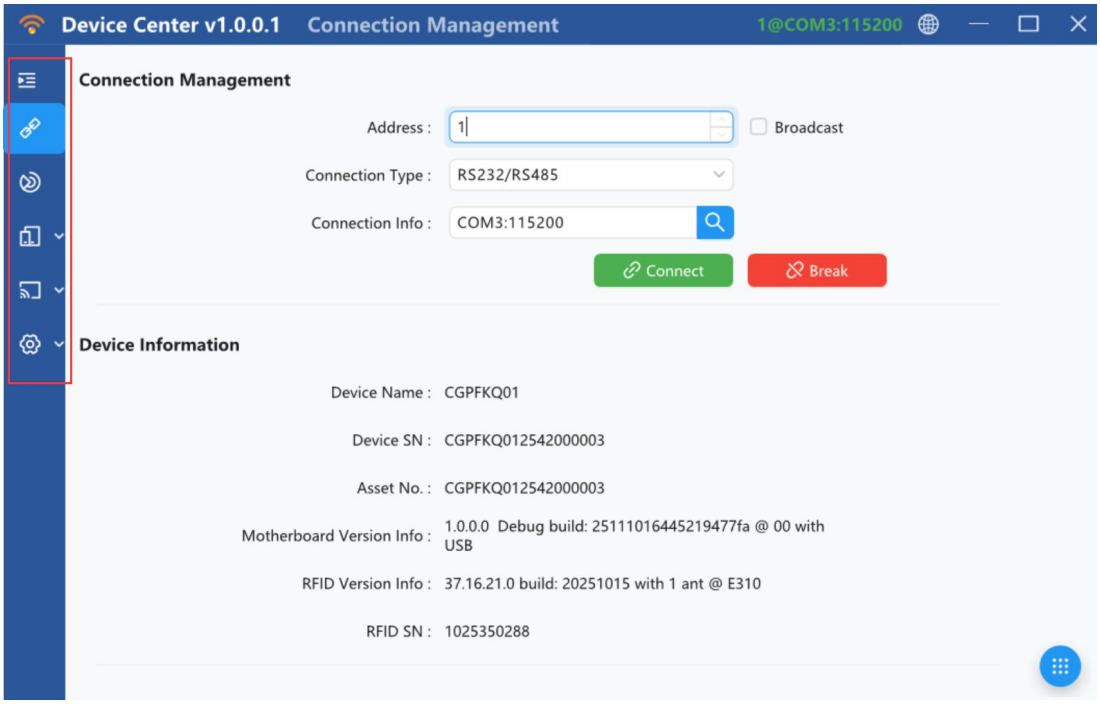
3、Click the "Stop" button to stop the inventory operation.

5. Interface Description

5.1. Interface Zones

The software interface mainly consists of a menu bar and a functional area. Users can click various menus on the menu bar to select and enter different functions within the functional area.

The menu bar has a collapsible function. Users can click the collapse/expand button to make it semi-hidden or fully pop out. The semi-hidden state is shown below:



5.2. Menu Description

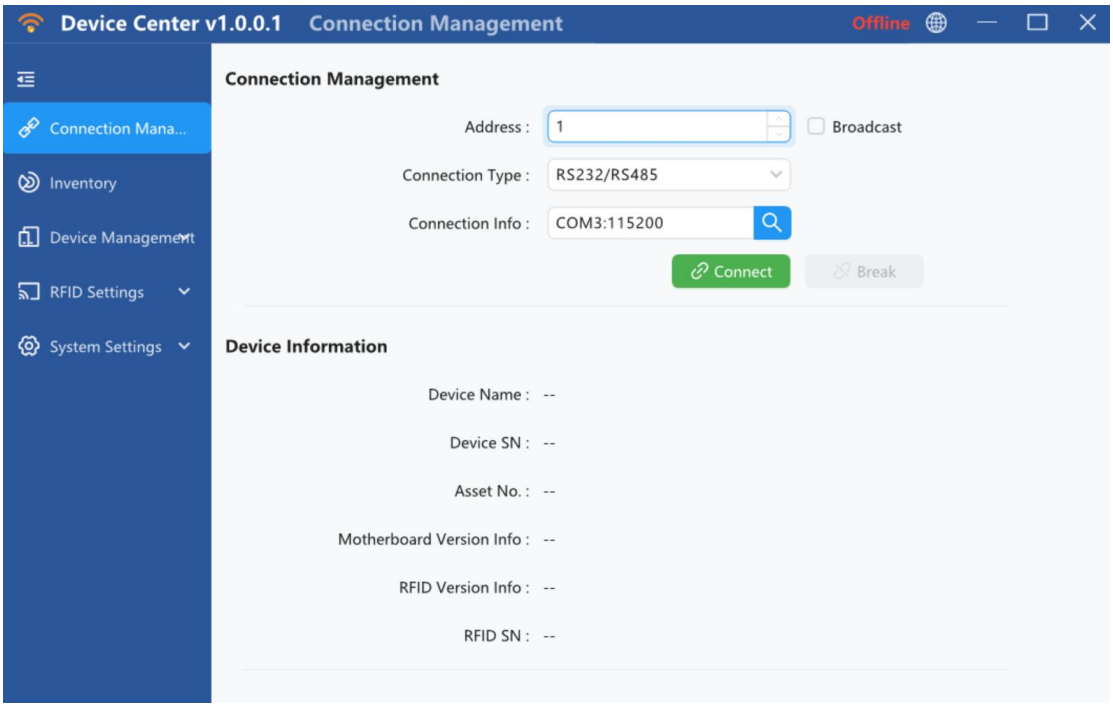
Primary Menu	Secondary Menu	Function Description
Connection Management		Connect to the device and display basic device information.
Inventory		RFID tag inventory, reading, writing, and other operations.
Device Management	Communication Settings	Query and set device address, RS232 parameters, RS485 parameters, etc.
	Network Settings	Query and set Ethernet network parameters and working mode.
	Wireless Settings	Query and set WiFi connection parameters, network parameters, working mode, and scan for hotspots.
	Buzzer & Indicator Light	Query and set buzzer parameters, communication indicator parameters, and tag indicator parameters.
	GPIO Settings	Query and set GPIO mode, level, events.
	Other Settings	Query and set device time, boot events.
RFID Settings	Antenna Settings	Query and set RFID antenna power.
	RFID Settings	Query and set RFID idle parameters, protocol parameters.
	UHF Settings	Query and set UHF working frequency, protocol parameters, and algorithm parameters.
	EAS Settings	Query and set device operations for EAS.
	Output Settings	Query and set device Wiegand output parameters, keyboard output parameters, and special format output parameters.
System Settings	System Settings	Query and set parameters such as parameter persistence.
	Log	View and set communication messages and events.

6. Software Functions

All operations on the device must first establish a device connection.

6.1. Connection Management

The Connection Management page is used to configure device connection parameters and establish device connections. Additionally, this page can read basic device information, including the device serial number and version. The Connection Management page is shown below:



The Connection Management page is also equipped with a set of floating action buttons. Users can click the button to expand or collapse them. These buttons cover functions such as device restart, device reset, command debugging, and device upgrade.

6.1.1. Connecting to a Device

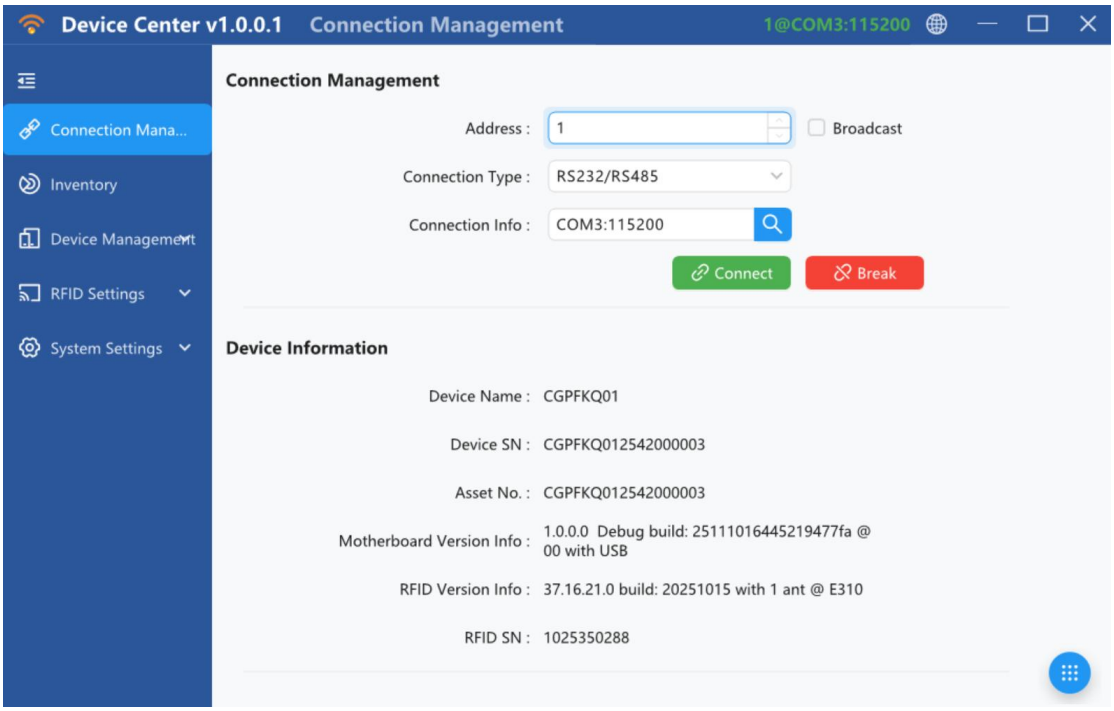
When connecting to a device, select the correct connection method and accurately fill in the communication address and connection information. The communication address value range is 1 to 250. If the reader/writer's communication address is uncertain, the broadcast communication address can be used for communication.

Supported connection methods include RS232/RS485 and TCP. The format of the connection information is as follows:

- RS232/RS485：[Serial Port: Baud Rate]，e.g., COM1：115200
- TCP：[IP：Port]，e.g., 192.168.1.201：8090

After entering the correct connection information, click the "Connect" button

to connect to the device. After a successful connection, the software will automatically query the device's basic information, as shown below:



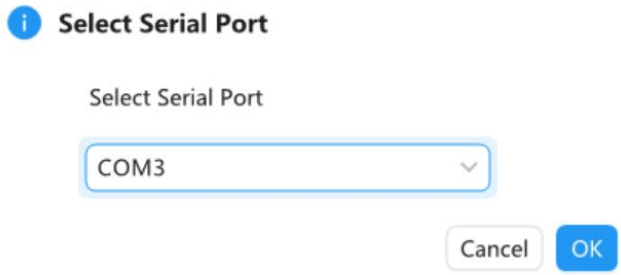
Device information content can be copied by double-clicking with the left mouse button.

When the user is unsure of the device's corresponding serial port number and baud rate, or does not know the device's IP address, they can perform a search by clicking the "Search" button next to the connection information box.

Clicking the "Search" button will cause the software to pop up a search interface for serial ports or network ports based on the selected connection method.

6.1.1.1. Search Serial Ports

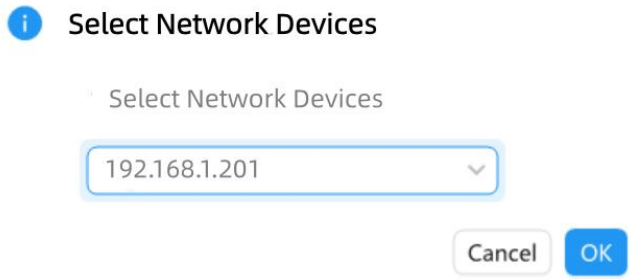
The Search Serial Ports interface is as follows:



After the user selects a serial port and clicks the OK button, the software will use adaptive baud rate technology to establish a connection with the device.

6.1.1.2. Search Network Devices

The Search Network Devices interface is as follows:



After the user selects an IP address, the software will establish a connection with the device through the default port.

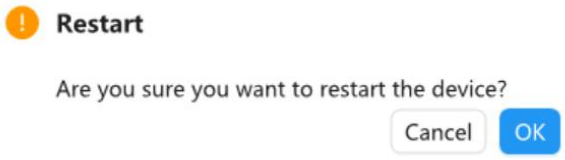
6.1.2. Disconnect the Device

The user can click the "Disconnect" button

to immediately disconnect the device.。

6.1.3. Device Restart

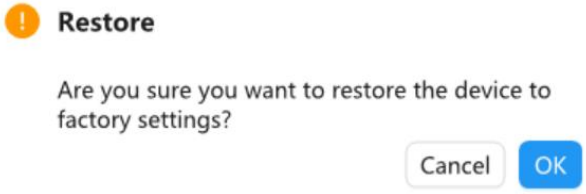
Click the floating action button to perform a device restart. After clicking, the software will first pop up a confirmation prompt to reconfirm with the user whether to perform the reset operation. The specific interface is shown below:



After confirmation, the software will send a reset command to the device; if Cancel is clicked, the current reset operation is abandoned.

6.1.4. Device Restore

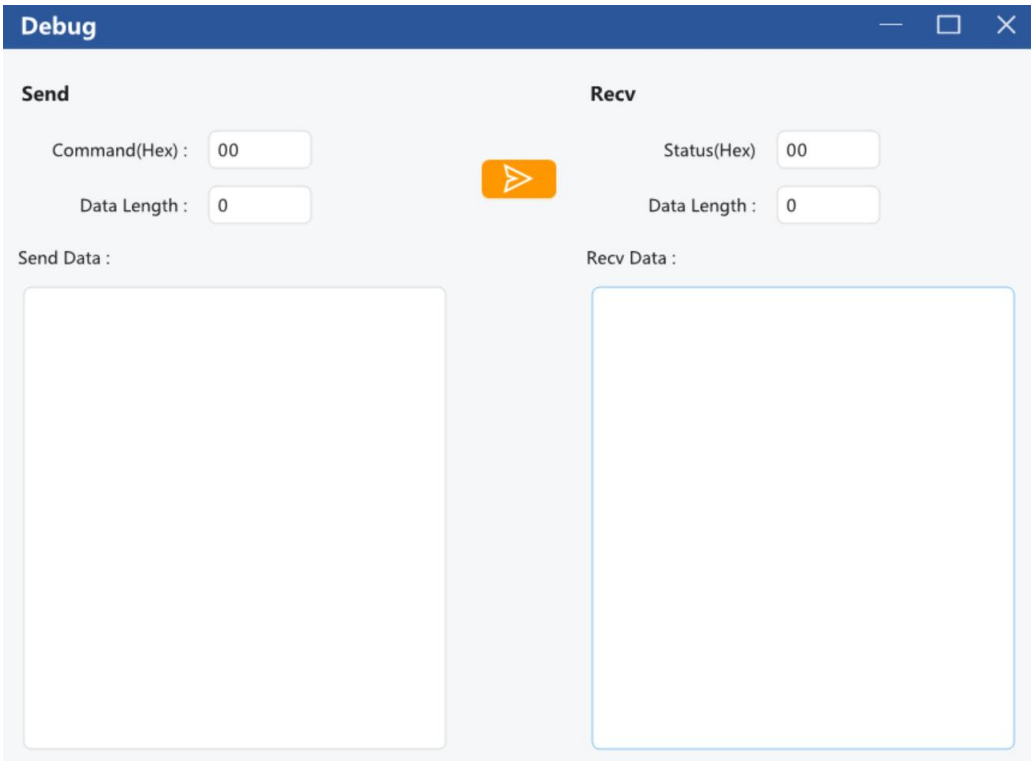
Click the floating action button to perform a device restore. After clicking, the software will first pop up a confirmation prompt to reconfirm with the user whether to perform the restore operation. The specific interface is shown below:



After confirm, the software will send a restore command to the device; if Cancel is clicked, the current reset operation is abandoned.

6.1.5. Command Debugging

Click the floating action button to open the Command Debugging interface. The specific interface is shown below:

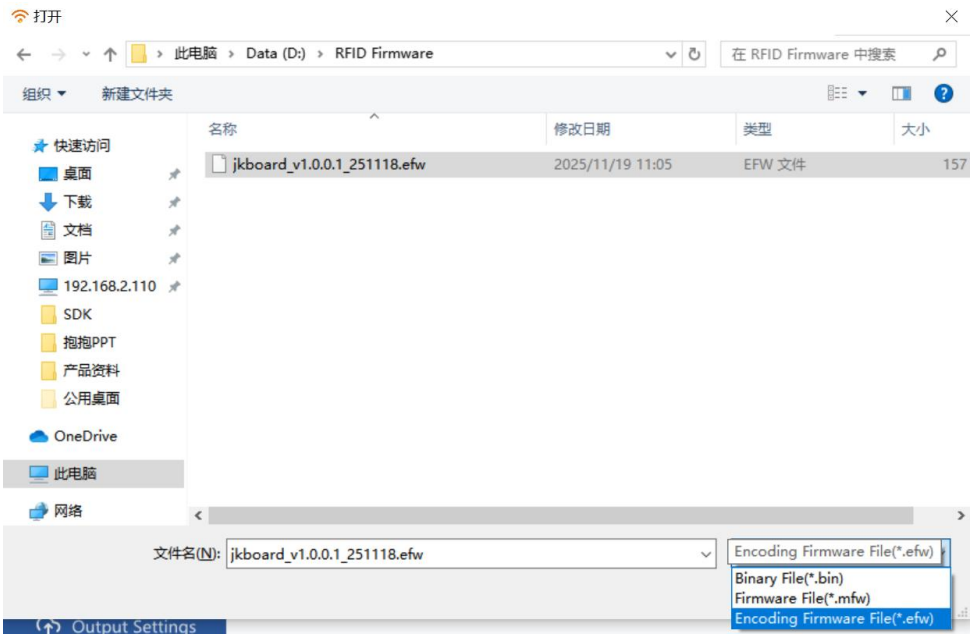


The left side of the debugging interface is the Send Area, where the command and data to be sent can be entered. The software will automatically calculate the data length and checksum. Click the "Send" button to send the command to the device.

The right side of the debugging interface is the Receive Area. When the device responds to the sent command, the software will populate the received response status code and data into the right-side interface; if no response data is received after sending the command and a timeout occurs, the Receive Data area will indicate a command timeout.

6.1.6. Device Upgrade

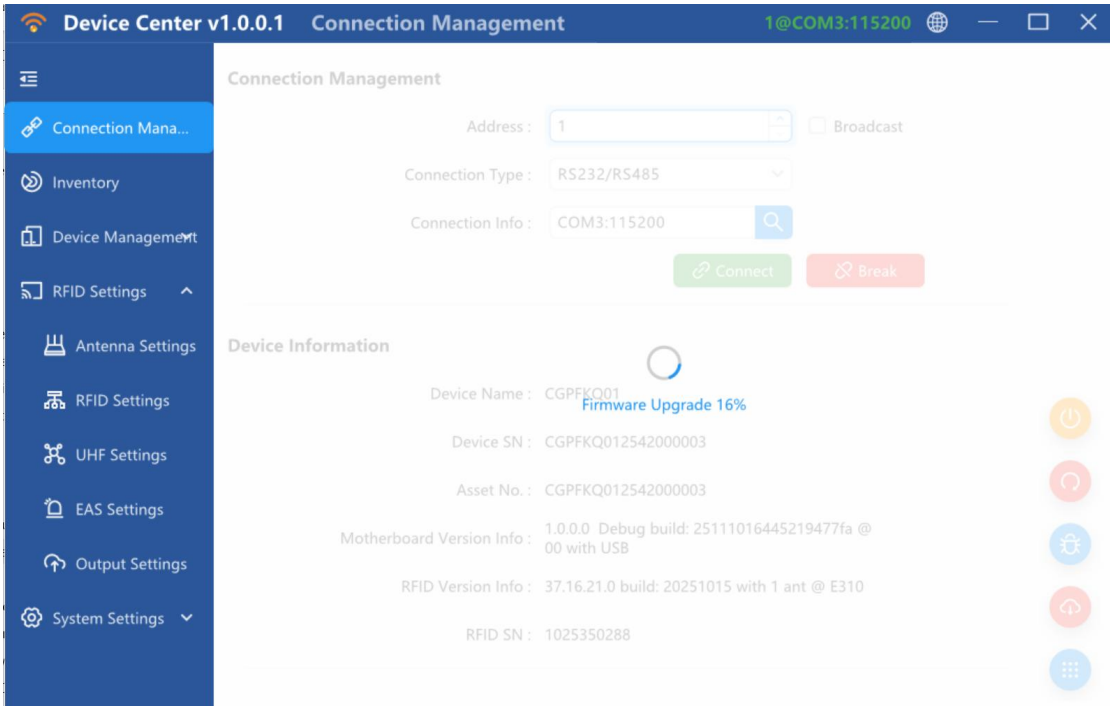
Click the floating action button to enter the Device Upgrade interface. Upon entry, the system will first pop up a file selection dialog for the user to choose the upgrade file. Supported file formats include bin, mfw, and efw.



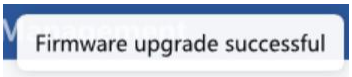
After the user selects the file, click the "Open" button

, the software will first pop up a confirmation prompt to reconfirm with the user whether to perform the upgrade operation. The specific interface is shown below:

Click "Confirm" to start the upgrade process; click "Cancel" to abandon the current upgrade operation. The upgrade process is shown below:

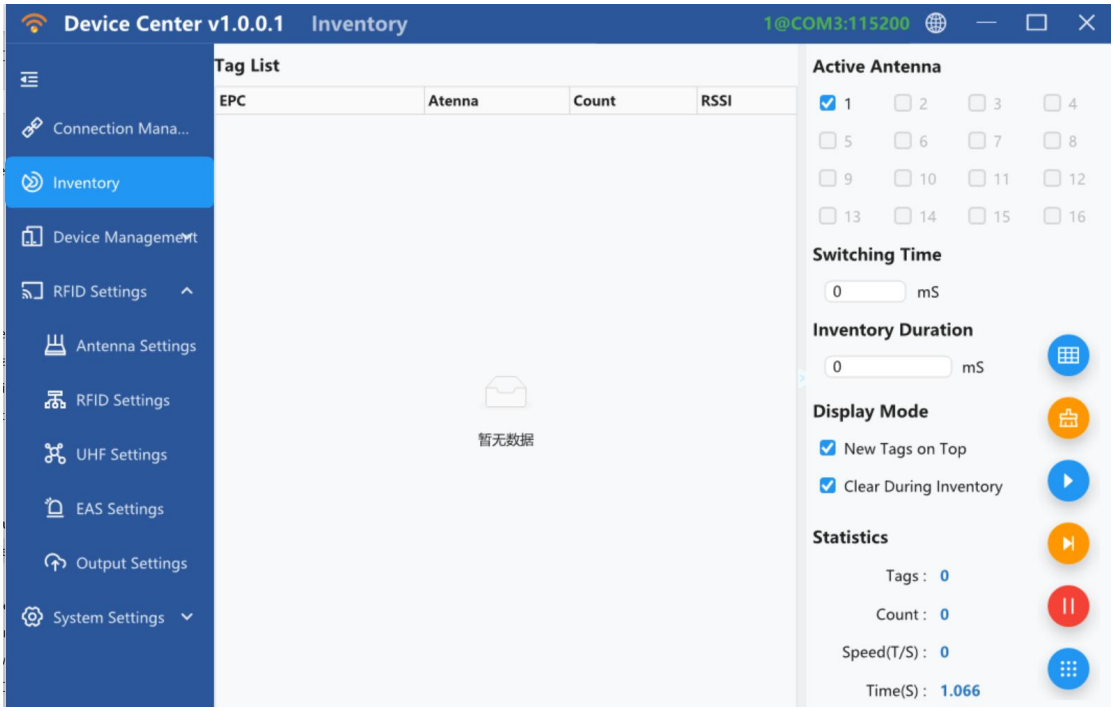


After the upgrade is completed, the top of the page will display the upgrade result prompt, as shown below:



6.2. Iventory

The Inventory page is used for RFID tag inventory, reading, writing, and other operations. It is used to test RFID functionality and performance. The Inventory page is shown below:



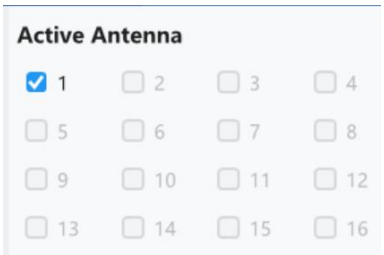
The Inventory page is also equipped with a set of floating action buttons. Users can click the button to expand or collapse them. These buttons cover functions such as Column Selection, Clear, Inventory, Advanced Inventory, Stop, and other operations.

6.2.1. Iventory Parameters

Inventory parameters are located on the right side of the page, covering Working Antenna, Antenna Switching Time, Inventory Duration, Display Mode, and Statistics.

6.2.1.1. Active Antenna

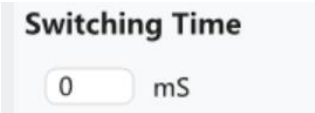
The Active Antenna parameter is used to select the antennas used in the inventory operation. Be sure to check only the required antennas.



Antenna power settings can be configured through the Antenna Configuration page.

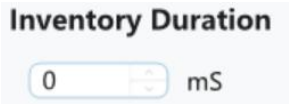
6.2.1.2. Antenna Swithing Time

The Antenna Switching Time is used to set the maximum dwell time for a single antenna in multi-antenna polling mode. The configurable range is 50 to 60000 ms. If configured as 0, it means it is managed automatically by the device.



6.2.1.3. Inventory Duration

The inventory operation can be either continuous or set for a specific duration. When the preset time is reached, the device will automatically stop the inventory. The inventory duration setting range is 50 to 60000 ms. If configured as 0, it means continuous inventory until the device receives a stop command.

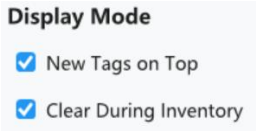


6.2.1.4. Display Mode

The Display Mode is used to set how tags are displayed, with the following options:

- New Tags on Top: When new tag data is inventoried, checking this option places the new tag data at the top for display; if unchecked, the data is inserted at the bottom for display.

- Clear on Inventory Start: After checking this option, the software will automatically clear the tag list when starting an inventory.

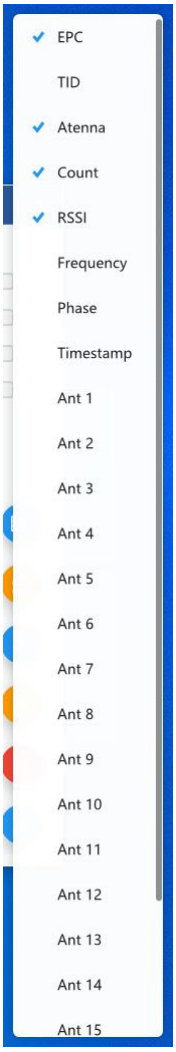


6.2.1.5. Statistics

Statistics are used to display the number of inventoried tags, the total number of tag reads, real-time inventory speed, and inventory time, as shown below:

6.2.2. Column Selection

Click the floating action button to perform the Column Selection operation. After clicking, the software will first pop up a selection menu. The specific interface is shown below:



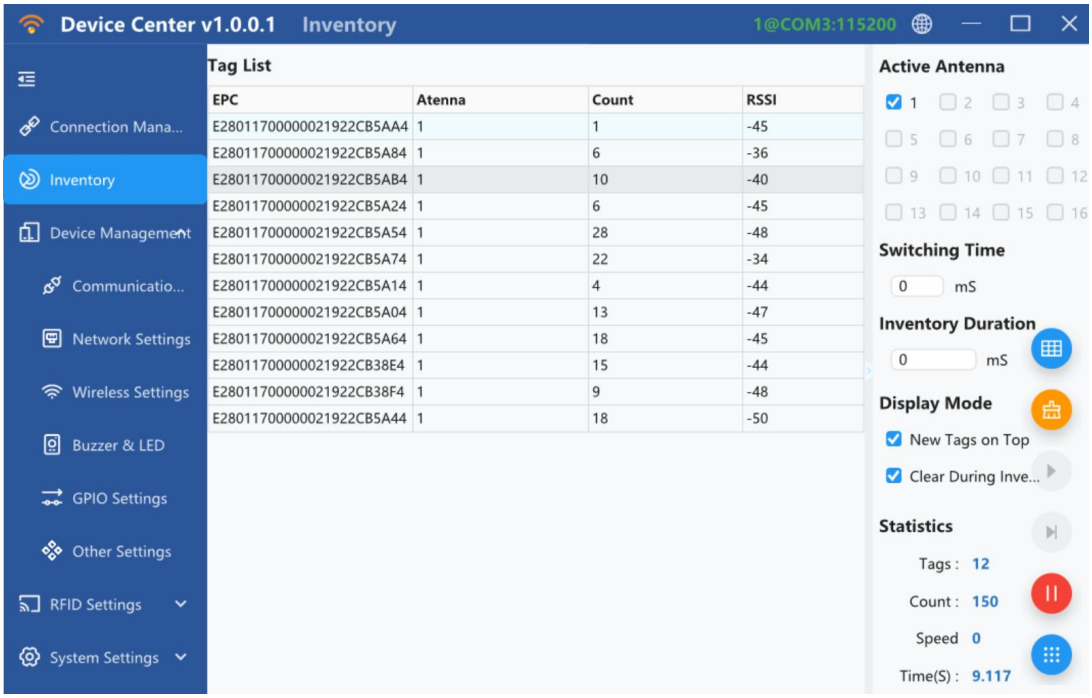
Users can check or uncheck menu items, and the corresponding tag data columns will be shown or hidden accordingly.

6.2.3. Clear

Click the floating action button to clear the tag list.

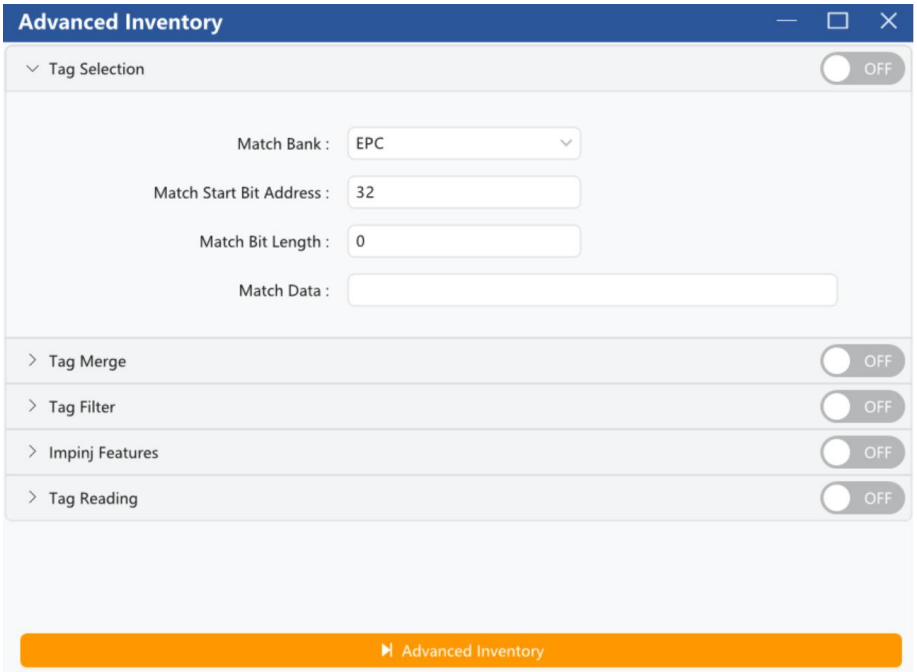
6.2.4. Iventory

Click the floating action button to perform the inventory operation. After starting the inventory, the inventoried tag data will be displayed in the tag list, as shown below:



6.2.5. Advanced Inventory

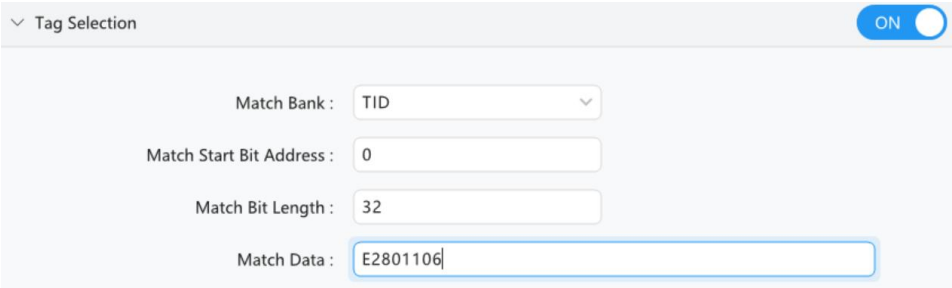
Click the floating action button to perform the Advanced Inventory operation. After clicking, the software will first pop up a dialog box for the user to configure advanced inventory parameters. The specific interface is shown below:



After configuring the advanced inventory parameters, click the "Advanced Inventory" button to start the inventory.

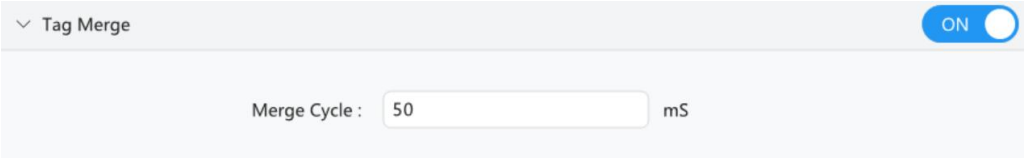
6.2.5.1. Tag Selection

The Tag Selection parameter is used to match specified tags for inventory. After the user configures the tag selection parameters, they need to enable the Tag Selection function for it to take effect. The configuration interface is shown below:



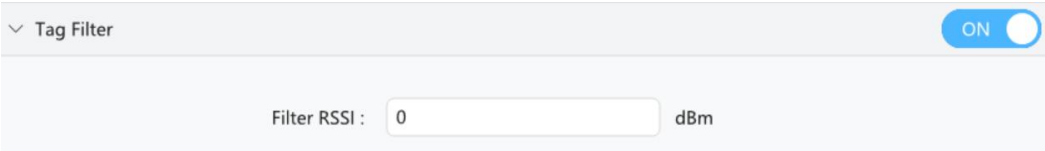
6.2.5.2. Tag Merge

Tag Merge is used to configure the device to merge tag data over time before uploading. After the user configures the tag merge parameters, they need to enable the Tag Merge function for it to take effect. The configuration interface is shown below:



6.2.5.3. Tag Filtering

Tag Filtering is used to configure the device to filter tag data based on signal strength before uploading. After the user configures the tag filtering parameters, they need to enable the Tag Filtering function for it to take effect. The configuration interface is shown below:

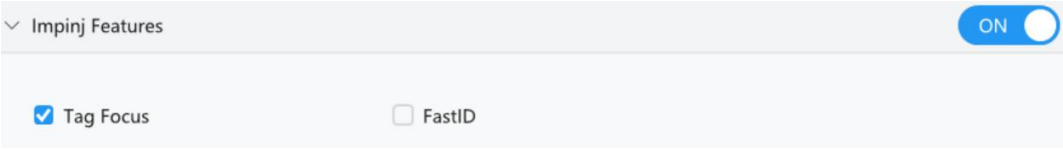


6.2.5.4. Impinj Tag Features

Impinj Tag Features are used to enable related functions for operating Impinj tags, including:

- TagFocus: Requires Session=1 and Target=A. After enabling, the tag responds only once before the S1 power loss.
- FastID: Requires Session=1 and Target=A. After enabling, during inventory, the TID is merged into the EPC and returned. The device will disassemble it and upload it in the standard format.

After configuring the corresponding Impinj tag feature parameters, they need to be enabled to take effect. The configuration interface is shown below:

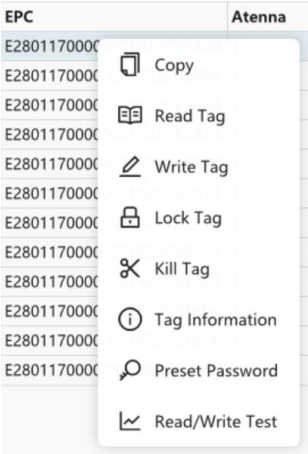


6.2.6. Stop

Click the floating action button to perform the Stop operation.

6.2.7. UHF Tag Operations

After inventorying UHF tags, operations such as reading and writing can be performed on the tags. Click on the tag data and right-click the mouse to open the tag operation menu, as shown below:

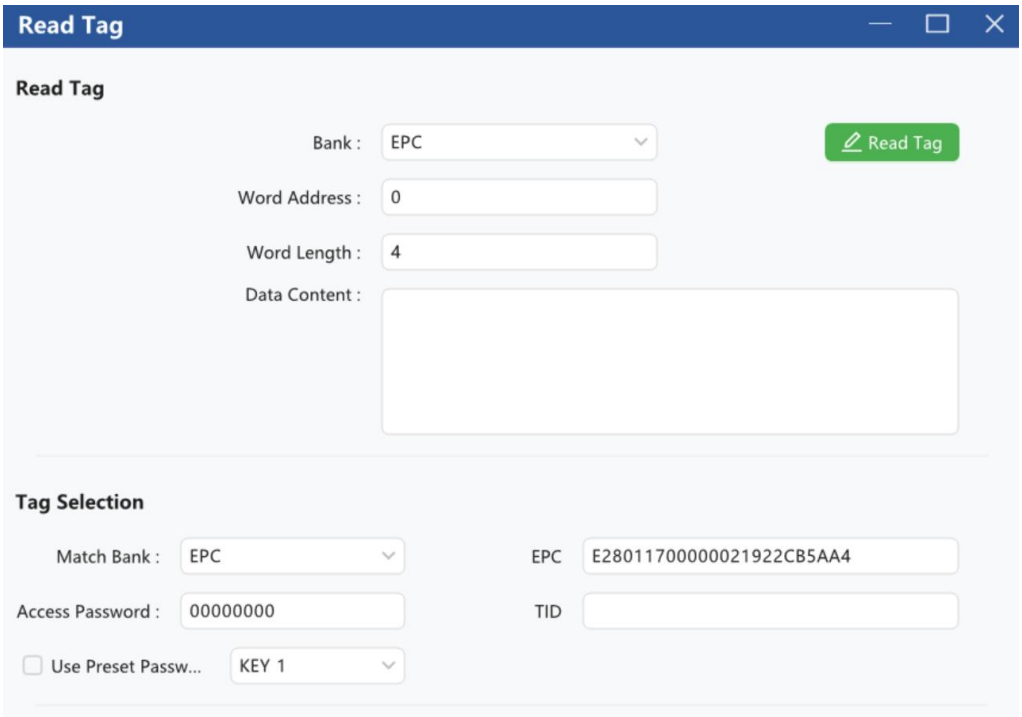


6.2.7.1. Copy

Clicking the "Copy" menu copies the content of the cell where the mouse was clicked to the clipboard.

6.2.7.2. Read Tag

Clicking the "Read Tag" menu will pop up the Read Tag interface, as shown below:



The Read Tag parameters are used to configure the memory bank to be read, the starting word address, and the number of words to read during the tag reading process.

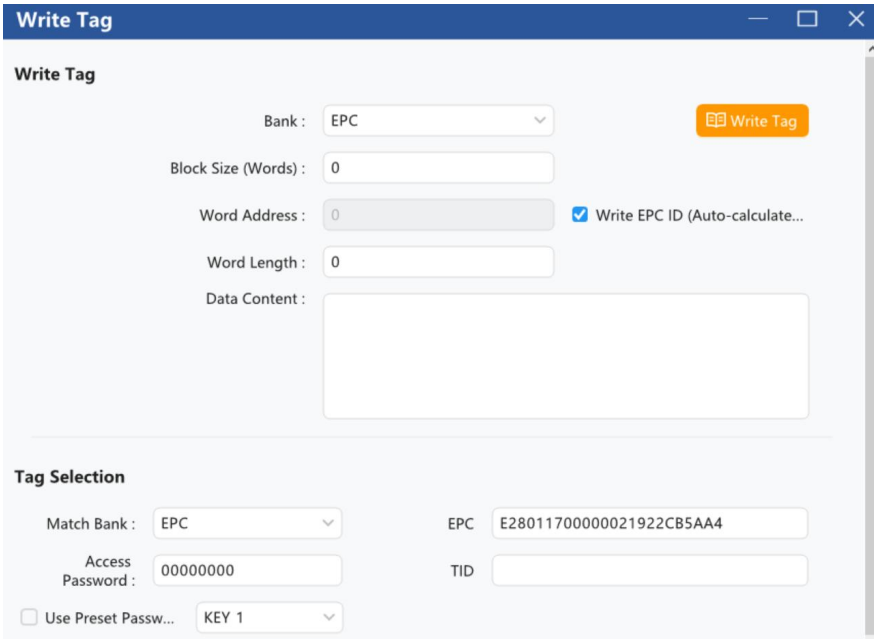
The Tag Selection parameters are used to configure the memory bank to match during tag selection and the tag access password. The tag access password can optionally use a preset password. When checked, the device will automatically use the KEY stored in the device instead of the access password.

After configuring the parameters, click the "Read Tag" button

to perform the tag reading operation. If the read fails, a corresponding prompt message will be displayed.

6.2.7.3. Write Tag

Clicking the "Write Tag" menu will pop up the Write Tag interface, as shown below:



The Write Tag parameters are used to configure the memory bank to be written, block size (used for block write operations, set to 0 for normal write), the starting word address, and the data to be written during the tag writing process.

The Tag Selection parameters are used to configure the memory bank to match during tag selection and the tag access password. The tag access password can optionally use a preset password. When checked, the device will automatically use the KEY stored in the device instead of the access password.

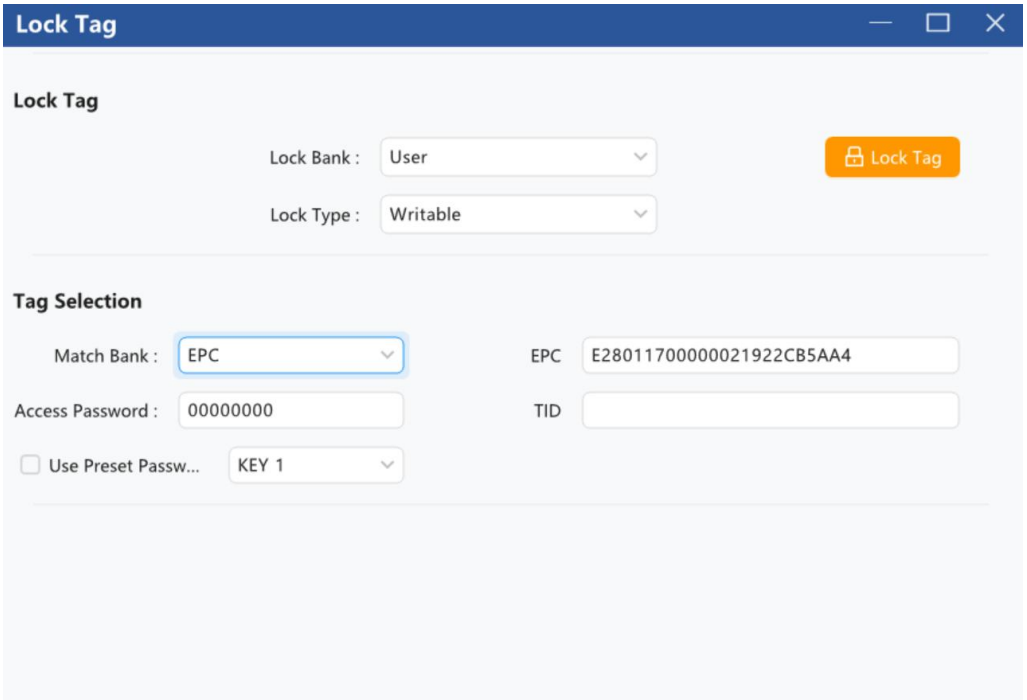
When the write memory bank is EPC, the EPC ID option can be checked. When checked, the software will automatically calculate the tag PC value and write it to the corresponding address.

After configuring the parameters, click the "Write Tag" button

to perform the tag writing operation. If the write fails, a corresponding prompt message will be displayed.

6.2.7.4. Lock Tag

Clicking the "Lock Tag" menu will pop up the Lock Tag interface, as shown below:



The Lock Tag parameters are used to configure the area on which to perform the lock operation and the type of lock operation during the tag locking process.

The Tag Selection parameters are used to configure the memory bank to match during tag selection and the tag access password. The tag access password can optionally use a preset password. When checked, the device will automatically use the KEY stored in the device instead of the access password.

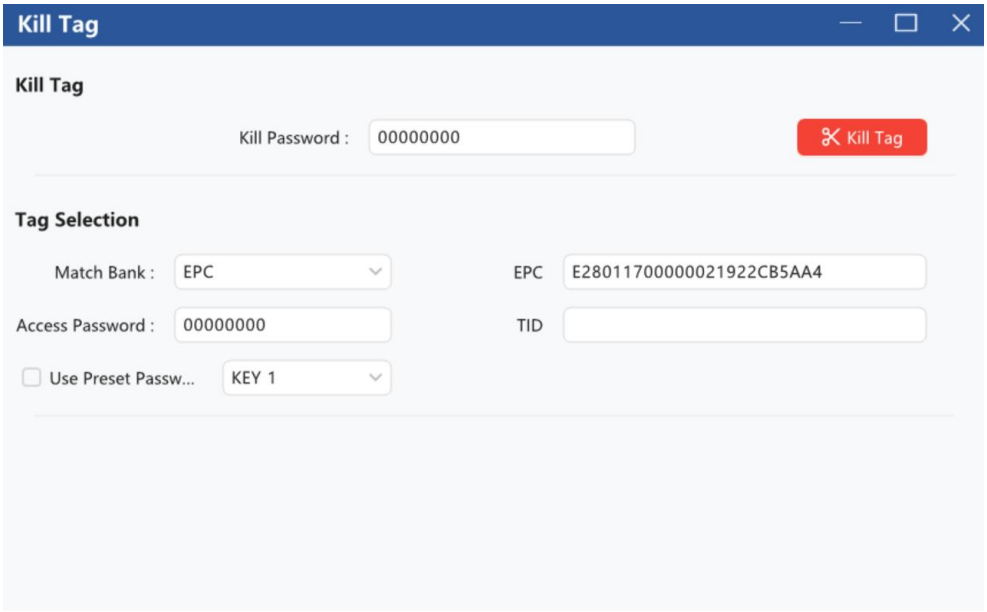
After configuring the parameters, click the "Lock Tag" button

to perform the tag lock operation. If the lock fails, a corresponding prompt message will be displayed.

Note: Before performing the tag lock operation, the tag's access password (Reserved bank: word address 2, word length 2) needs to be modified to a value other than all zeros.

6.2.7.5. Kill Tag

Clicking the "Kill Tag" menu will pop up the Kill Tag interface, as shown below:



The Kill Tag parameter is used to configure the kill password used to kill the tag.

The Tag Selection parameters are used to configure the memory bank to match during tag selection.

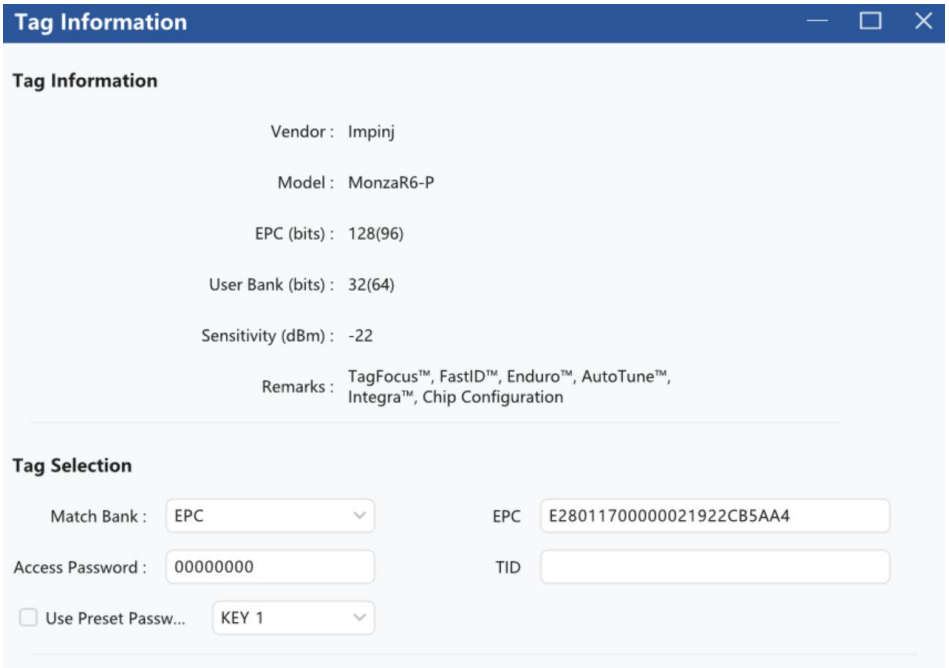
After configuring the parameters, click the "Kill Tag" button

to perform the tag kill operation. If the kill fails, a corresponding prompt message will be displayed.

Note: Before performing the tag kill operation, the tag's kill password (Reserved bank: word address 0, word length 2) needs to be modified to a value other than all zeros.

6.2.7.6. Tag Information

Clicking the "Tag Information" menu will pop up the Tag Information interface, as shown below:

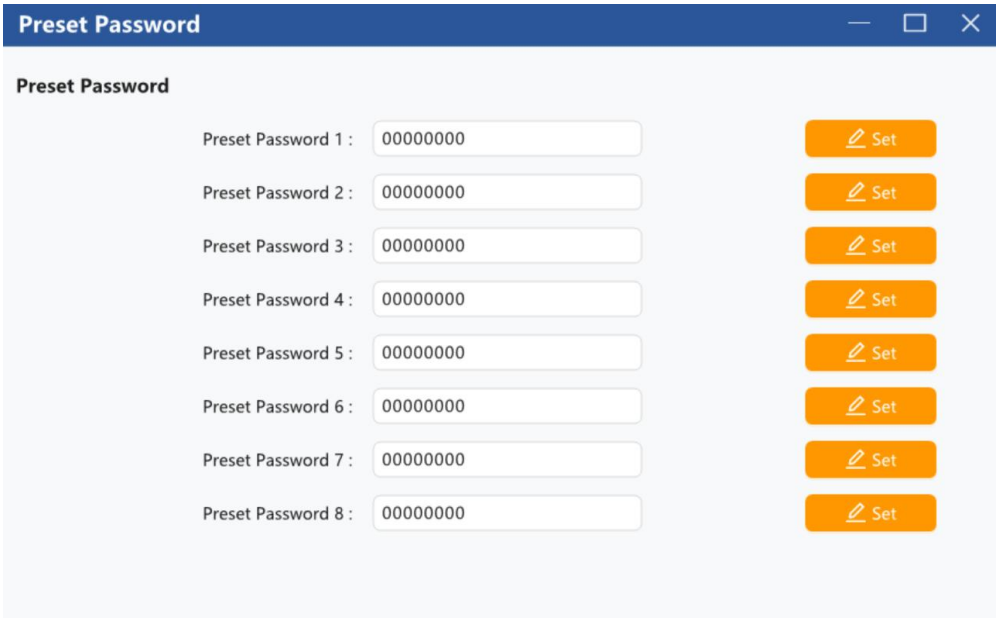


The Tag Information interface displays information such as the tag manufacturer, model, memory bank sizes, sensitivity, etc.

Note: To accurately display tag information, TID needs to be read during inventory.

6.2.7.7. Preset Password Management

Clicking the "Preset Password Management" menu will pop up the Preset Password Management interface, as shown below:



The Preset Password Management interface allows modification of 8 sets of preset passwords.

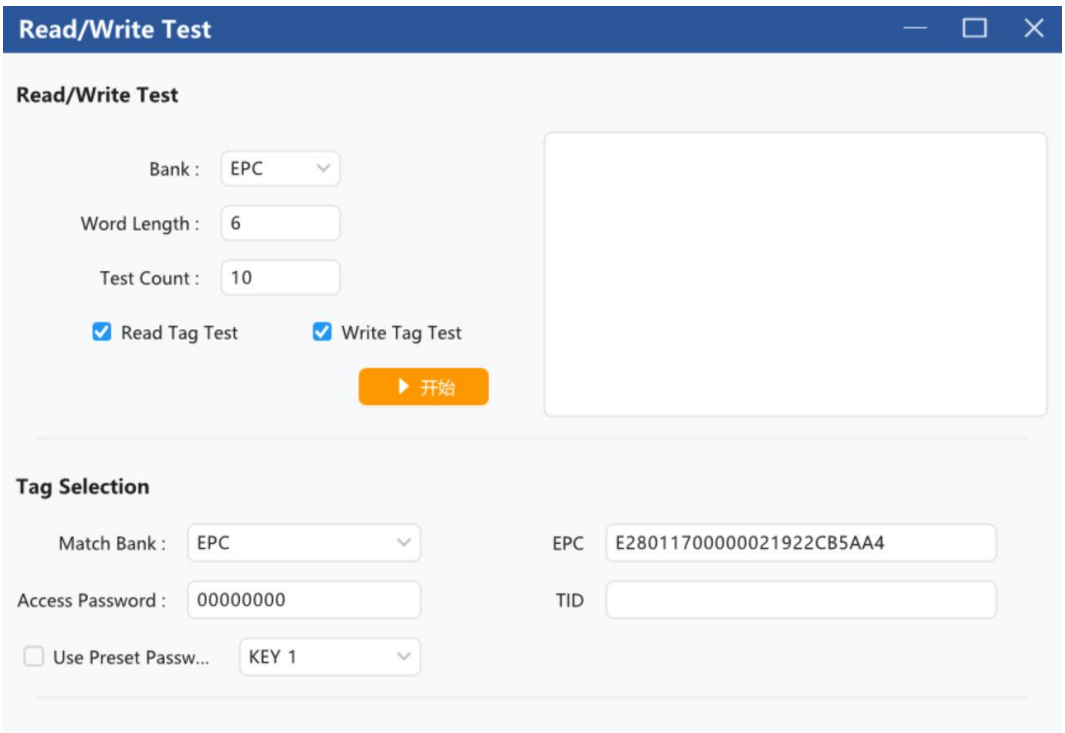
After modifying the corresponding preset password, click the "Set" button

next to it, and the software will set the preset password into the device.

Note: Whether the device supports preset passwords and how many groups it supports need to be checked in the corresponding specification sheet.

6.2.7.8. Read/Write Test

Clicking the "Read/Write Test" menu will pop up the Read/Write Test interface, as shown below:



Read/Write Test parameters include the memory bank to be tested, word length, number of test cycles, whether to perform read tag tests, whether to perform write tag tests, etc.

The Tag Selection parameters are used to configure the memory bank to match during tag selection and the tag access password. The tag access password can optionally use a preset password. When checked, the device will automatically use the KEY stored in the device instead of the access password.

After configuring the parameters, click the "Start" button

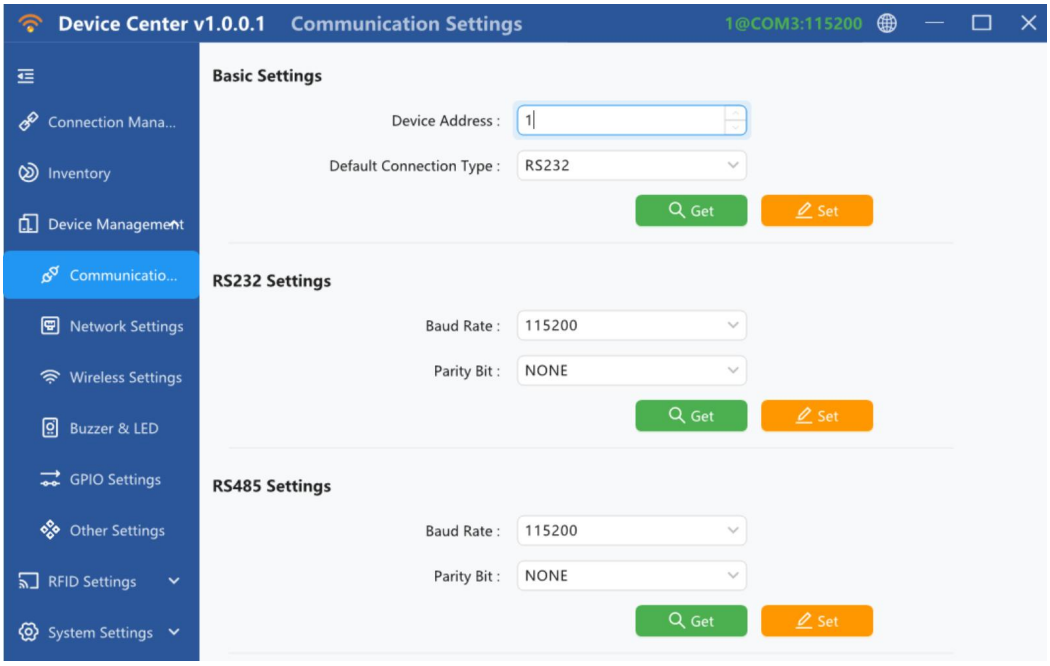
to perform the continuous tag read/write operation; Click the "Stop" button

to stop the continuous tag read/write operation. The read/write test results are displayed in the text box on the right.

6.3. Device Management

6.3.1. Communication Settings

The Communication Settings page covers multiple functions such as Basic Settings (e.g., device address), RS232 Settings, and RS485 Settings. The layout of this page is shown below:



6.3.1.1. Basic Settings

The Basic Settings interface is shown below:

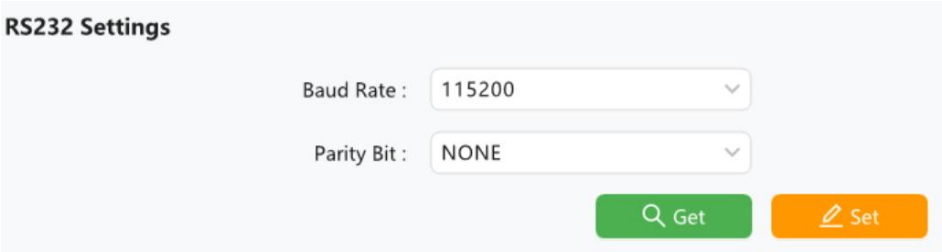
Basic Settings parameters include the following:

- Device Address: The address used for device communication, ranging from 1 to 250.
- Default Connection Method: The communication connection method automatically adopted by the device after power-on. Options can be set to RS232, RS485, TCP, etc.

Click the "Get" button to query the Basic Settings parameters. Click the "Set" button to modify the device's Basic Settings parameters.

6.3.1.2. RS232 Settings

The RS232 Settings interface is shown below:



RS232 Settings parameters include the following:

- Baud Rate: The baud rate used for RS232 communication. Supports 9600, 19200, 115200, 230400, and 460800. Default is 115200.
- Parity Bit: The parity bit used for RS232 communication. Supports None, Odd, and Even. Default is None.

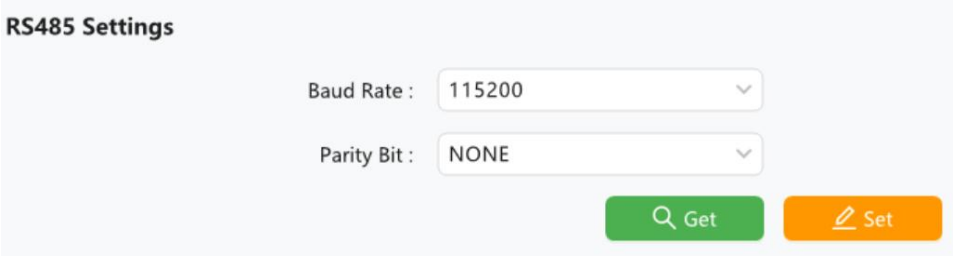
Click the "Get" button

to query the RS232 Settings parameters , Click the "Set" button

to modify the device's RS232 Settings parameters.

6.3.1.3. RS485 Settings

The RS485 Settings interface is shown below:



RS485 Settings parameters include the following:

- Baud Rate: The baud rate used for RS485 communication. Supports 9600, 19200, 115200, 230400, and 460800. Default is 115200.
- Parity Bit: The parity bit used for RS485 communication. Supports None, Odd, and Even. Default is None.

Click the "Get" button

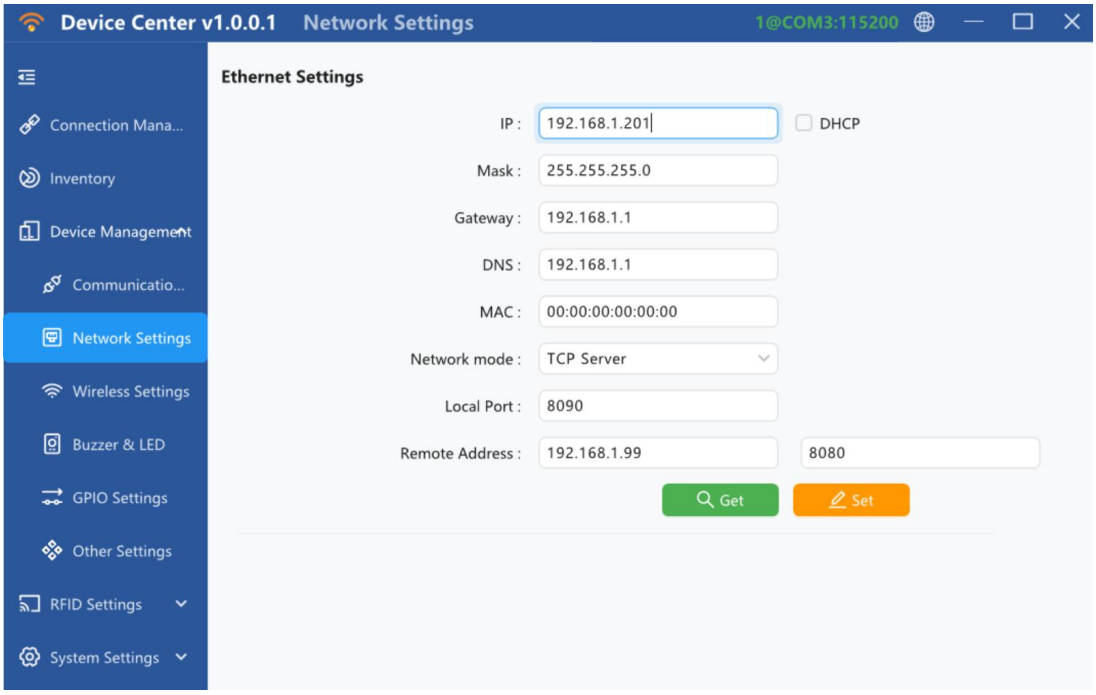
to query the RS485 Settings parameters; Click the "Set" button

to modify the device's RS485 Settings parameters.

6.3.2. Network Settings

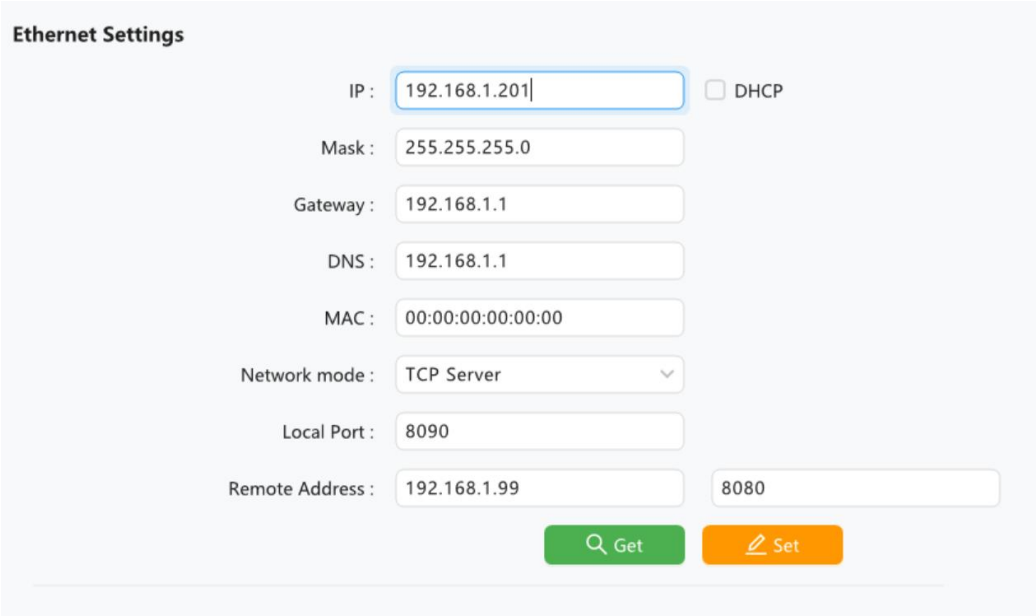
The Network Settings page covers Ethernet network parameters and working mode settings. The Network Settings page is shown

below:



6.3.2.1. Enthernet Settings

The Ethernet Settings interface is shown below:



Ethernet Settings parameters include the following:

- DHCP: Whether the Ethernet uses DHCP to obtain an IP address.
- IP Address: The IP address used by the Ethernet.
- Subnet Mask: The subnet mask used by the Ethernet.

- Gateway Address: The gateway address used by the Ethernet.
- DNS Address: The DNS server address used by the Ethernet.
- Working Mode: The working mode used by the Ethernet. Can be set to TCP Server, TCP Client, UDP Server, and UDP Client. UDP mode requires device support. Default is TCP Client.
- Local Port: The local port used for Ethernet TCP/UDP communication. Default is 8090.
- Remote Address: The remote address and port used for Ethernet TCP Client/UDP Client communication. Defaults are 192.168.1.99 and 8090 respectively.
- MAC Address: The physical address of the Ethernet.

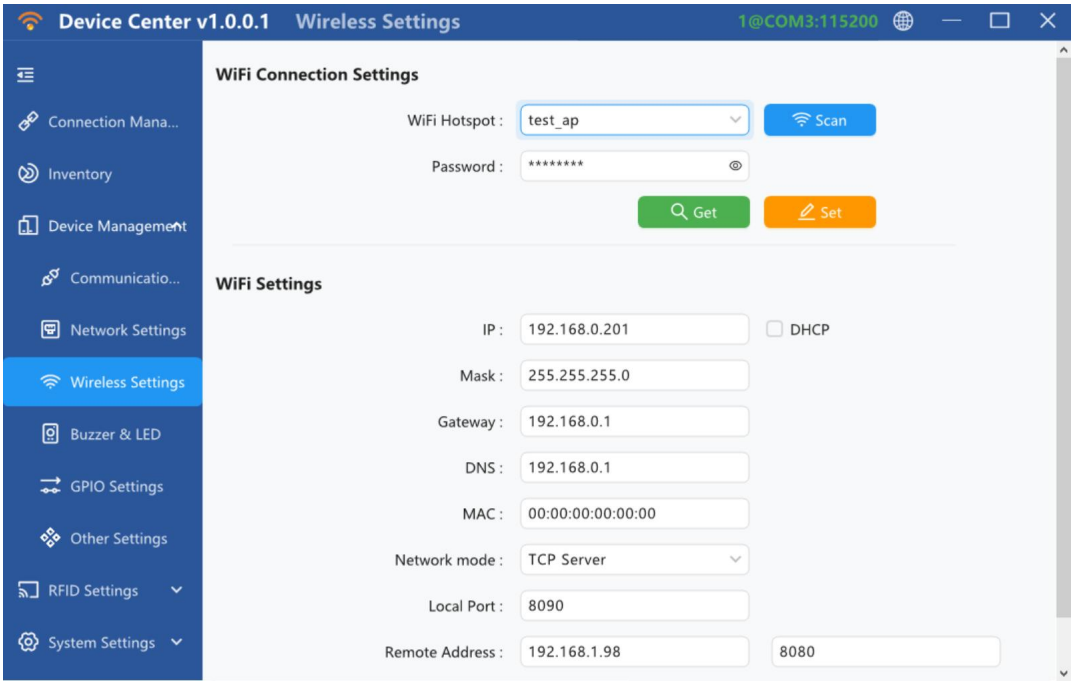
Click the "Get" button

to query the Ethernet Settings parameters, Click the "Set" button

to modify the device's Ethernet Settings parameters.

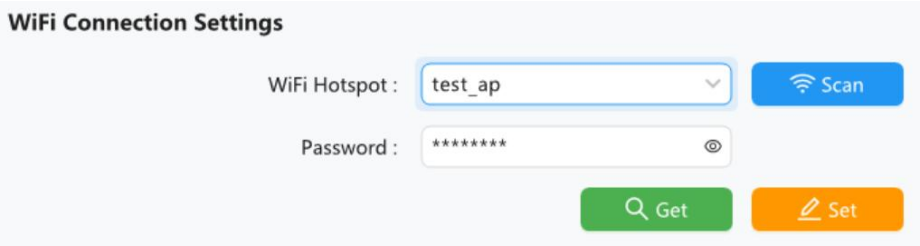
6.3.3. Wireless Settings

The Wireless Settings page covers WiFi network parameters, working mode settings, and WiFi connection parameter settings. The Wireless Settings page is shown below:



6.3.3.1. WiFi Connection Settings

The Wireless Settings page covers WiFi network parameters, working mode settings, and WiFi connection parameter settings. The Wireless Settings page is shown below:



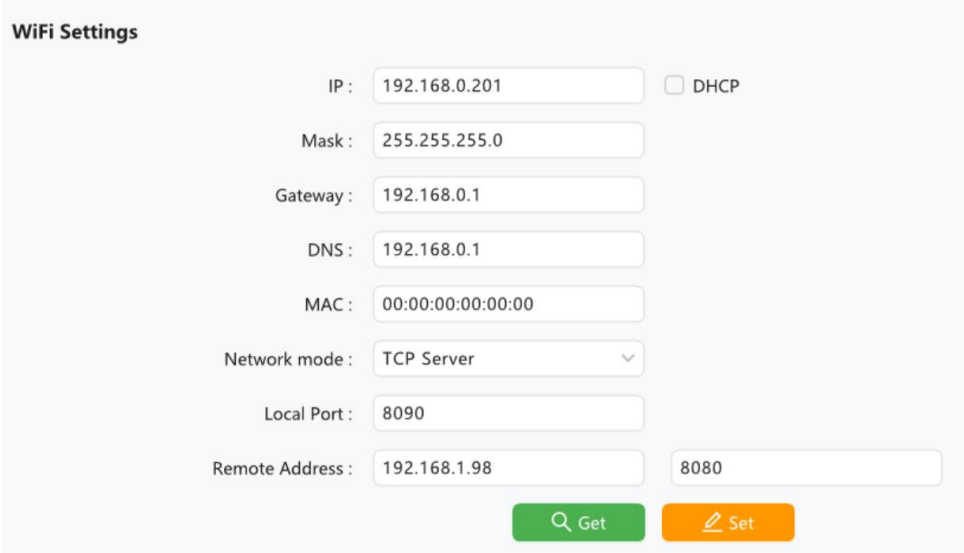
WiFi Connection Settings parameters include the following:

- WiFi Hotspot (SSID): The AP for the WiFi connection. Can be selected after scanning by clicking the "Scan" button
- Password: The password used to connect to the WiFi AP.

Click the "Get" button to query the WiFi Connection Settings parameters. Click the "Set" button to modify the device's WiFi Connection Settings parameters.

6.3.3.2. WiFi Settings

The WiFi Settings interface is shown below:



WiFi Settings parameters include the following:

- DHCP: Whether the WiFi uses DHCP to obtain an IP address.
- IP Address: The IP address used by the WiFi.
- Subnet Mask: The subnet mask used by the WiFi.
- Gateway Address: The gateway address used by the WiFi.
- DNS Address: The DNS server address used by the WiFi.
- Working Mode: The working mode used by the WiFi. Can be set to TCP Server, TCP Client, UDP Server, and UDP Client. UDP mode requires device support. Default is TCP Client.
- Local Port: The local port used for WiFi TCP/UDP communication. Default is 8090.
- Remote Address: The remote address and port used for WiFi TCP Client/UDP Client communication. Defaults are 192.168.1.99 and 8090 respectively.
- MAC Address: The physical address of the WiFi.

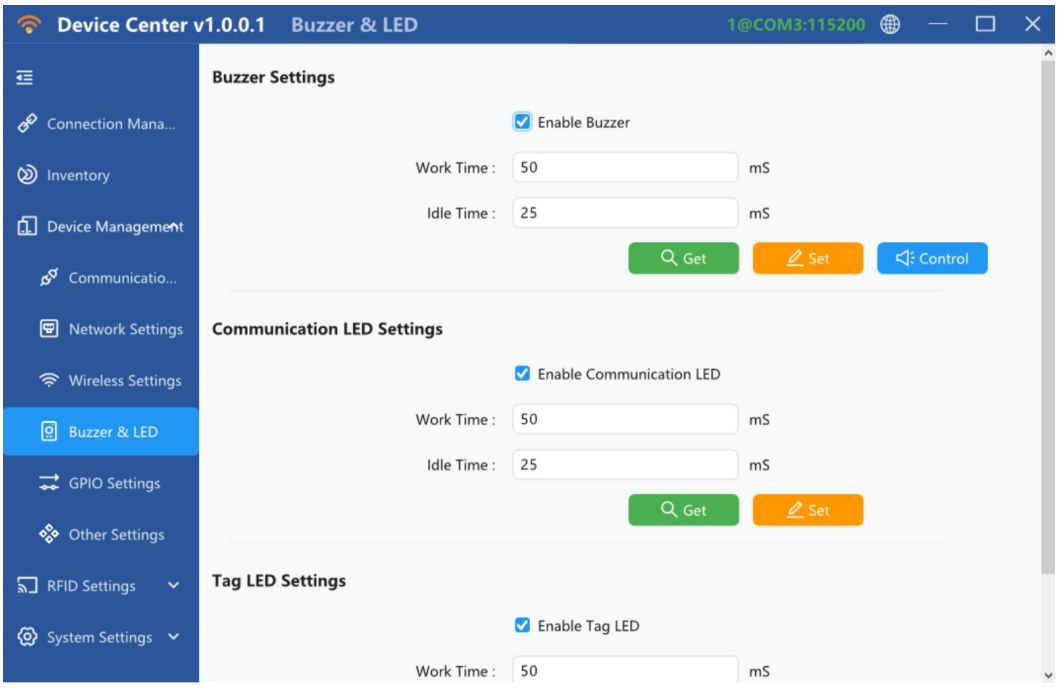
Click the "Get" button

to query the WiFi Settings parameters; Click the "Set" button

to modify the device's WiFi Settings parameters .

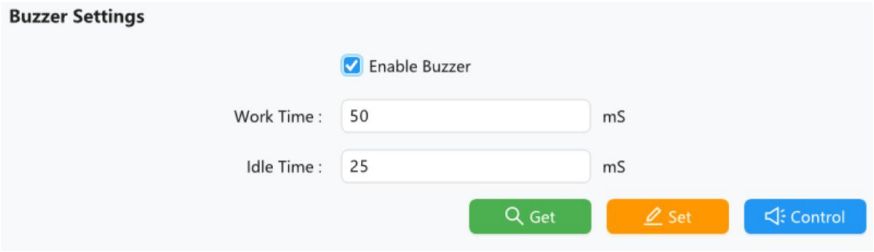
6.3.4. Buzzer & Indicator Light

The Buzzer & Indicator Light page covers Buzzer Settings, Communication Indicator Settings, and Tag Indicator Settings. The Buzzer & Indicator Light page is shown below:



6.3.4.1. Buzzer Settings

The device buzzer can be controlled by the host computer to sound, or it can sound when a tag is inventoried. The Buzzer Settings interface is shown below:



Buzzer Settings parameters include the following:

- Enable Buzzer: Whether to enable the buzzer. After enabling the buzzer, the device will sound once for each tag inventoried.
- Work Duration: The duration the buzzer sounds, in milliseconds. Default is 50.
- Idle Duration: The duration the buzzer is silent, i.e., the pause time between two consecutive sounds, in milliseconds. Default is 25.

Click the "Get" button

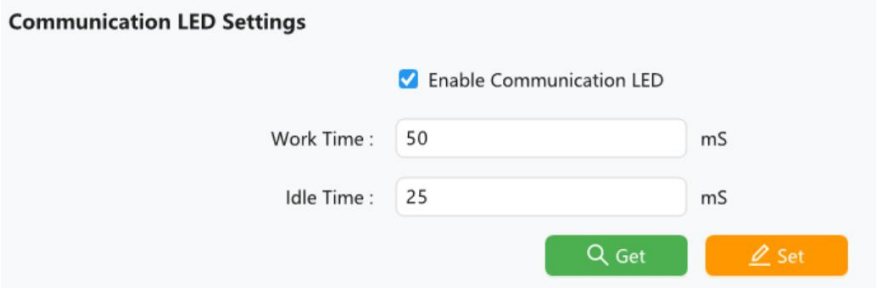
to query the Buzzer Settings parameters. Click the "Set" button

to modify the device's Buzzer Settings parameters. Click the "Control" button

to make the buzzer sound once.

6.3.4.2. Communication Indicator Settings

The Communication Indicator Settings interface is shown below:



Communication Indicator Settings parameters include the following:

- Enable Communication Indicator: Whether to enable the communication indicator. After enabling, the indicator light will illuminate once for each communication.
- Work Duration: The duration the communication indicator light is on, in milliseconds. Default is 50.
- Idle Duration: The duration the communication indicator light is off, i.e., the pause time between two consecutive illuminations, in milliseconds. Default is 25.

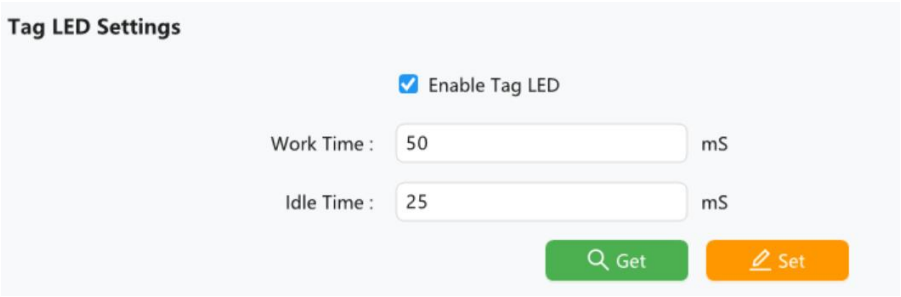
Click the "Get" button

to query the Communication Indicator Settings parameters ; Click the "Set" button

to modify the device's Communication Indicator Settings parameters.

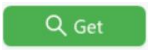
6.3.4.3. Tag Indicator Settings

The Tag Indicator Settings interface is shown below:



Tag Indicator Settings parameters include the following:

- Enable Tag Indicator: Whether to enable the tag indicator. After enabling, the indicator light will illuminate once for each tag inventoried.
- Work Duration: The duration the tag indicator light is on, in milliseconds. Default is 50.
- Idle Duration: The duration the tag indicator light is off, i.e., the pause time between two consecutive illuminations, in milliseconds. Default is 25.

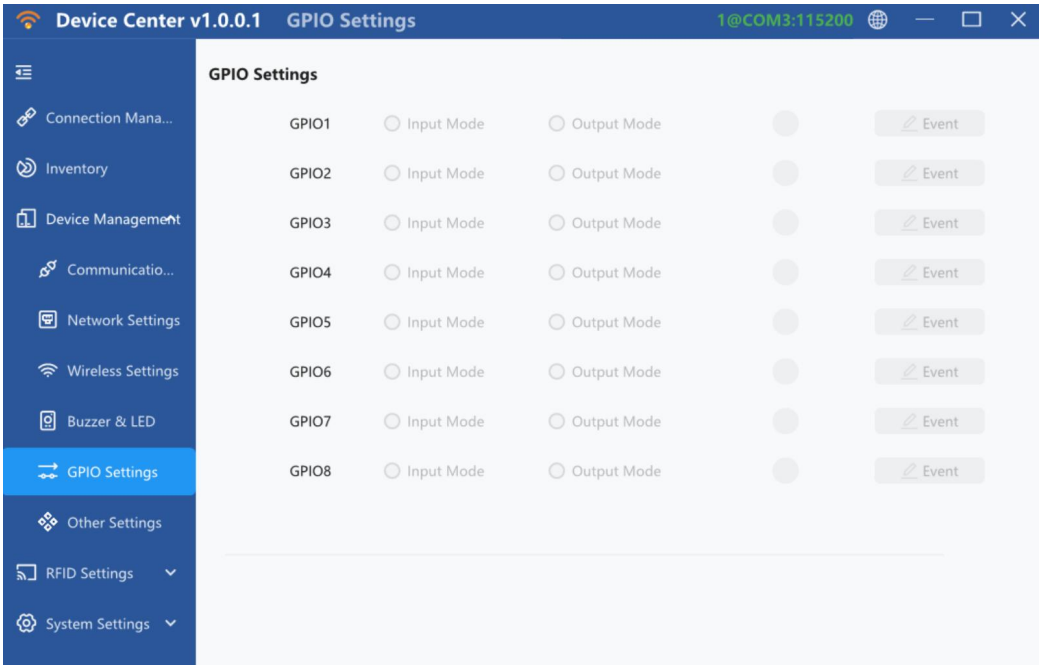
Click the "Get" button 

to query the Tag Indicator Settings parameters. Click the "Set" button

to modify the device's Tag Indicator Settings parameters .

6.3.5. GPIO Settings

The GPIO page covers GPIO Mode Settings, GPIO Level Settings & Query, and GPIO Event Configuration. The GPIO page is shown below:



6.3.5.1. GPIO Mode Configuration

GPIO can be configured as Input or Output mode. It can be switched immediately by clicking the Input Mode/Output Mode radio buttons, as shown below:

Note: Whether the GPIO mode can be modified needs to be checked in the device specification sheet.

6.3.5.2. GPIO Level Configuration and Query

For GPIO configured in Output mode, the output level can be set by clicking the button ;

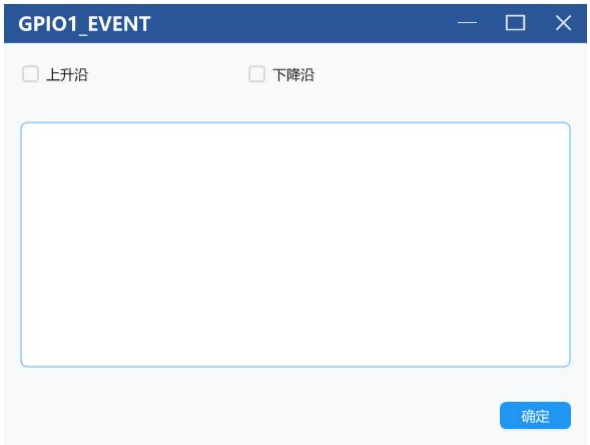
For GPIO configured in Input mode, the input level can be queried by clicking the button ;

Note: For devices that support GPIO change event reporting, they will actively upload changes in the level of GPIO in Input mode; the software will automatically update their GPIO level status.

6.3.5.3. GPIO Event Configuration

For GPIO in Input mode, the device can execute a GPIO event on its rising/falling edge. The GPIO Event Configuration interface can be popped up by clicking the "Event" button

as shown below:

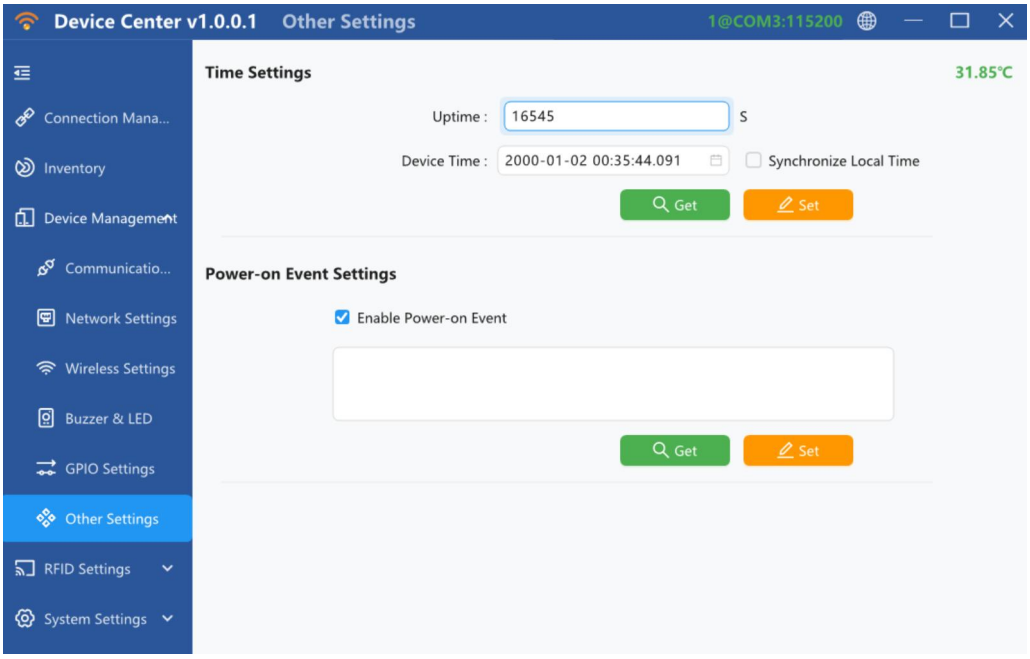


The user can select the trigger edge and configure the command (full command) to be executed when the event is triggered. Click OK to configure.

In the command input box, you can right-click the mouse to pop up a menu for quickly inputting common commands, as shown below:

6.3.6. Other Settings

The Other Settings page covers Time Settings and Boot Event Settings. The Other Settings interface is shown below:



6.3.6.1. Time Settings

The Time Settings interface is shown below:

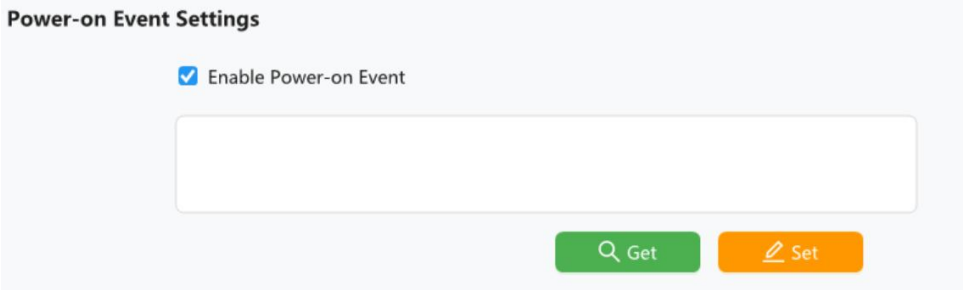
Time Settings parameters include the following:

- Uptime: The time elapsed since the device was powered on, in seconds.
- Device Time: The device's current time.
- Sync with PC Time: When checked, the device time will be synchronized with the PC time when setting the device time.

Click the "Get" button to query the Time Settings parameters. Click the "Set" button to modify the device's Time Settings parameters.

6.3.6.2. Power-on Event Settings

The Power-on Event Settings interface is shown below:

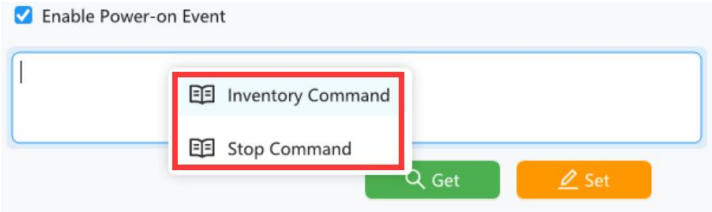


Power-on Event Settings parameters include the following:

- Enable Power-on Event: Whether to enable the boot event. When checked, the Power-on event command is triggered when the device starts up.
- Event Command: The command (full command) executed by the Power-on event.

Click the "Get" button to query the Power-on Event Settings parameters , Click the "Set" button to modify the device's Power-on Event Settings parameters.

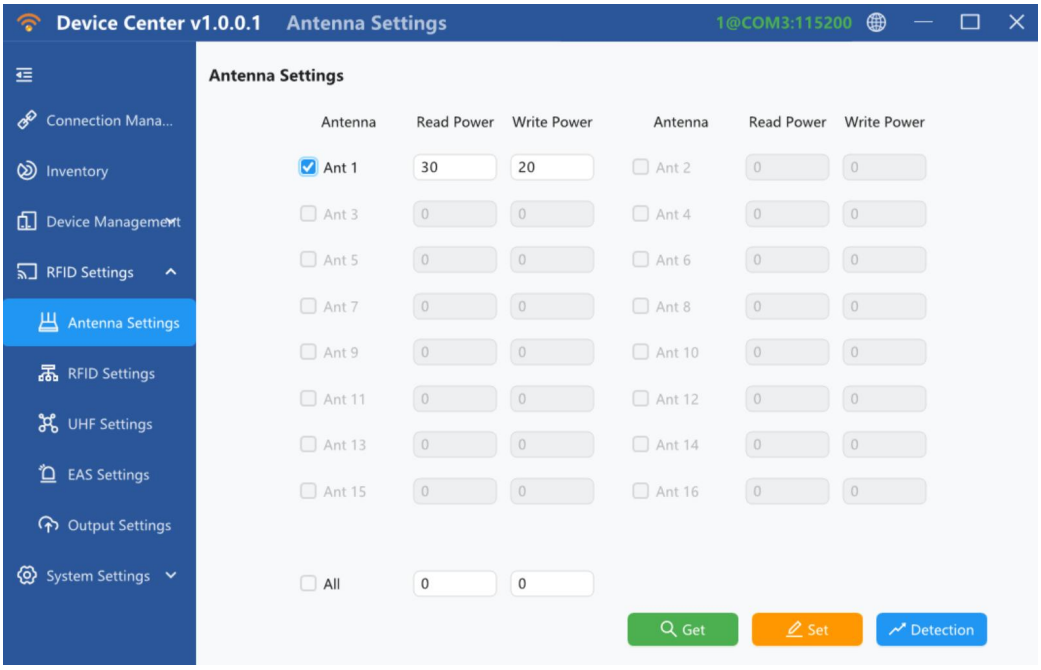
In the command input box, you can right-click the mouse to pop up a menu for quickly inputting common commands, as shown below:



6.4. RFID Settings

6.4.1. Antenna Settings

The Antenna Settings page covers the antenna's Read Power (for inventory, read tag) and Write Power (for write tag, lock tag, kill tag) settings. The Antenna Settings interface is shown below:



When users need to modify antenna power, they must check the corresponding antenna and then adjust its Read Power or Write Power value. The reader/writer power unit is dBm.

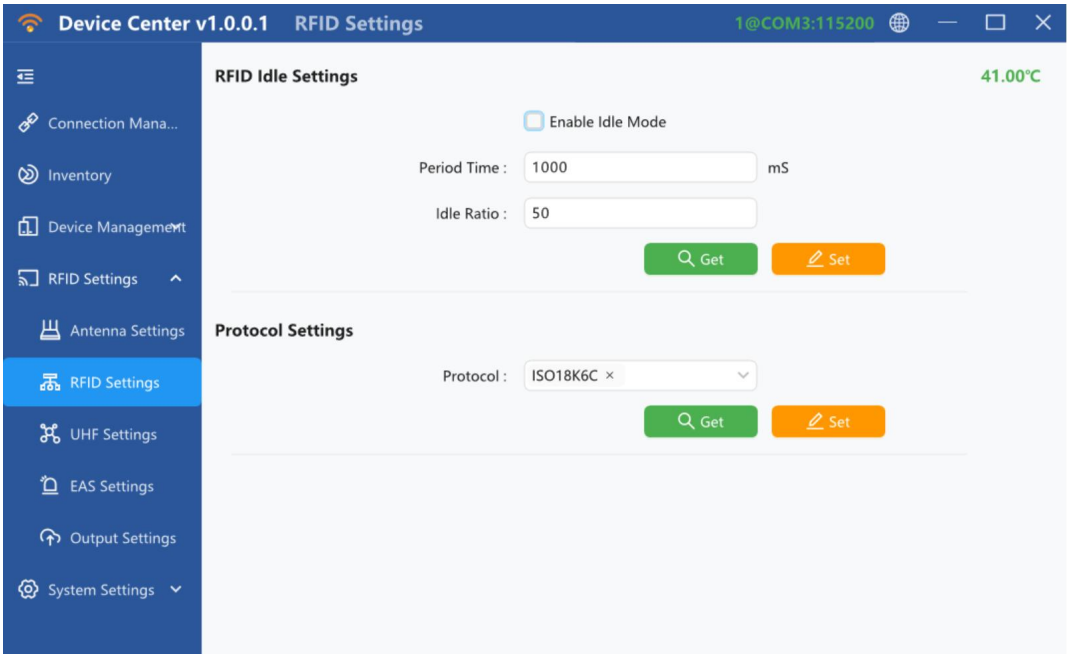
Click the "Get" button

to query the Antenna Settings parameters. Click the "Set" button

to modify the device's Antenna Settings parameters.

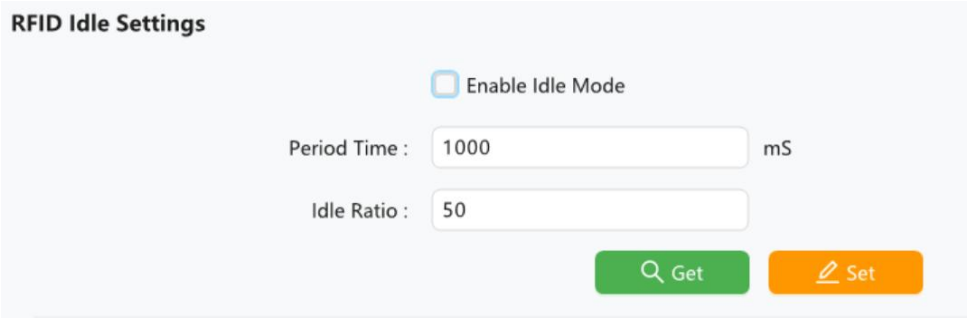
6.4.2. RFID Settings

The RFID Settings page covers functions such as RFID Idle Settings and RFID Protocol Settings. The RFID Settings page is shown below:



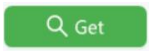
6.4.2.1. RFID Idle Settings

RFID Idle refers to the device periodically working and sleeping in proportion during the RFID inventory process. The RFID Idle Settings interface is shown below:



RFID Idle Settings parameters include the following:

- Enable Idle Mode: Whether to enable idle mode. After enabling idle mode, the device inventory will periodically execute work-sleep cycles in proportion.
- Cycle: The periodic time for the device to execute RFID inventory, in milliseconds. Value range is 50~60000.
- Idle Ratio: The proportion of sleep time within the cycle time during which the device executes RFID inventory.

Click the "Get" button 

to query the RFID Idle Settings parameters. Click the "Set" button

to modify the device's RFID Idle Settings parameters.

6.4.2.2. Protocal Settings

The RFID Protocol Settings are used to configure the protocol currently used for RFID work. The Protocol Settings interface is shown below:

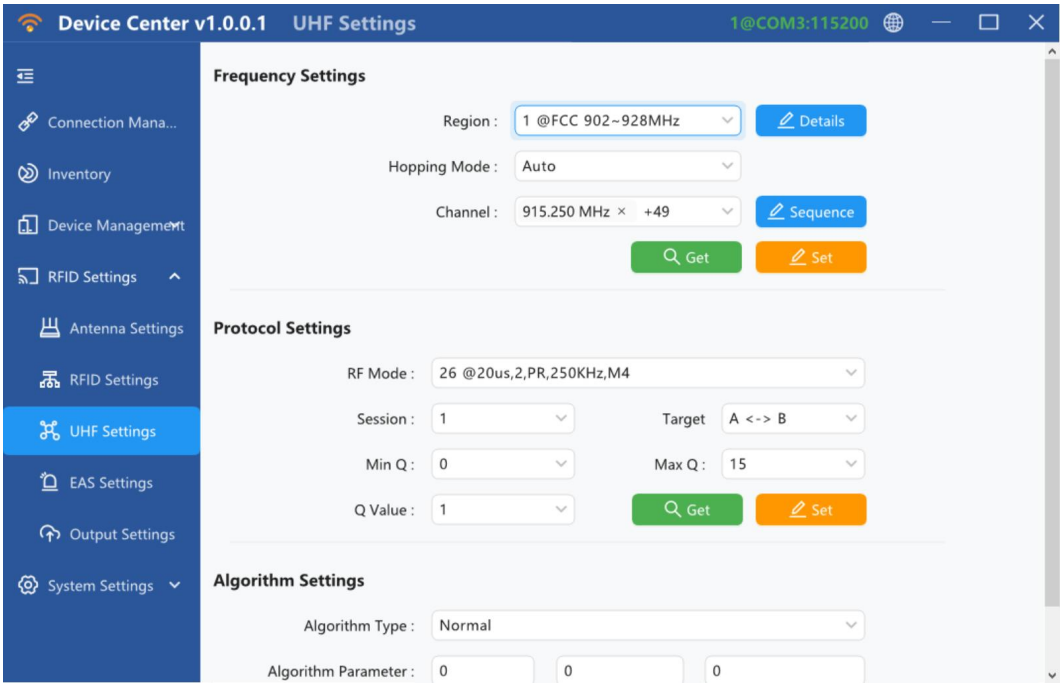
Click the "Get" button

to query the RFID Protocol Settings parameters. Click the "Set" button

to modify the device's RFID Protocol Settings parameters.

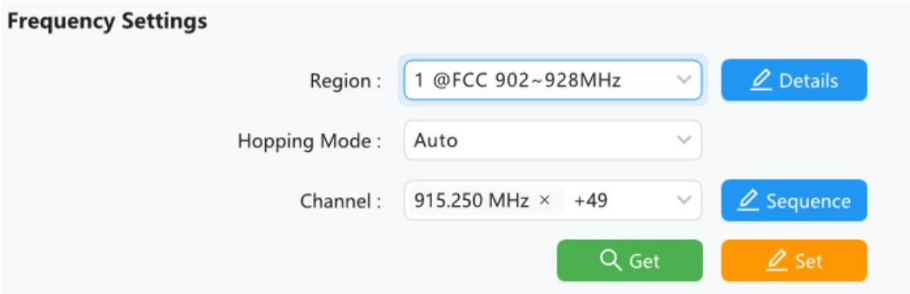
6.4.3. UHF Settings

The UHF Settings page covers functions such as Frequency Settings, Protocol Settings, and Algorithm Settings. The UHF Settings page is shown below:



6.4.3.1. Frequency Settings

The Frequency Settings interface is shown below:



Frequency Settings parameters include the following:

- Working Region: Select the working region that complies with local regulations, such as FCC, ETSI, CMIIT.
- Frequency Hopping Mode: Select the channel frequency hopping mode, which can be Automatic or Manual.
- Channels: When manual frequency hopping is selected, available channels and hopping sequence can be configured.

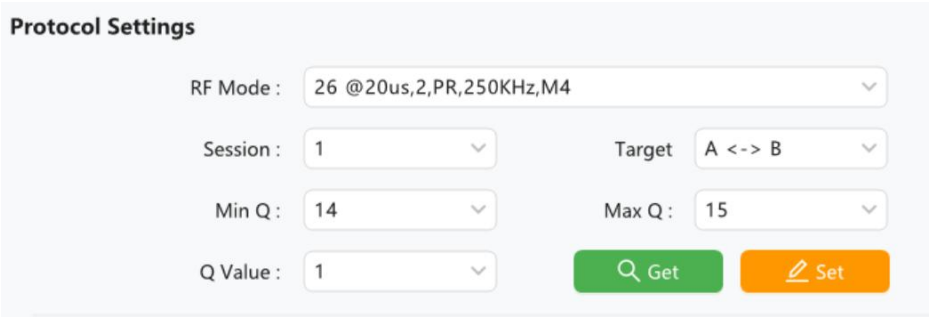
Click the "Get" button

to query the Frequency Settings parameters. Click the "Set" button

to modify the device's Frequency Settings parameters.

6.4.3.2. UHF Protocal Settings

The Protocol Settings interface is shown below:



Protocol Settings parameters include the following:

- Working Mode: The working mode determines the transmission rate and encoding method for communication between the device and tags. Different working modes affect the speed and sensitivity of tag inventory. Users can make adjustments according to their needs in practical applications. Note that different devices or modules may support different working options.
- Session: Session is a logical channel for interaction between the tag and the device. Each tag supports 4 independent Sessions (S0, S1, S2, S3), each maintaining a set of independent state flags.
- Target: Target is a binary parameter (A/B) specified when the device sends a Query command, used to select tags in a specific state flag within the current Session. The Target flips after inventory under specific conditions. A<->B and B<->A in Target are modes that automatically select the Target state in the Query command, differing in whether the initial query state uses A or B.
- Default Q Value: Q determines the number of response slots tags randomly choose from (Total slots = 2^Q). The Q value range is 0

to 15. A larger Q value is suitable for reading more tags; Q=0 is suitable for single-tag scenarios. The Default Q value is the Q value used by the device in its initial state during inventory.

- Minimum Q Value: The minimum Q value used by the dynamic Q algorithm. When Max Q, Min Q, and Default Q are set to the same value, the static Q algorithm is used.
- Maximum Q Value: The maximum Q value used by the dynamic Q algorithm. When Max Q, Min Q, and Default Q are set to the same value, the static Q algorithm is used.

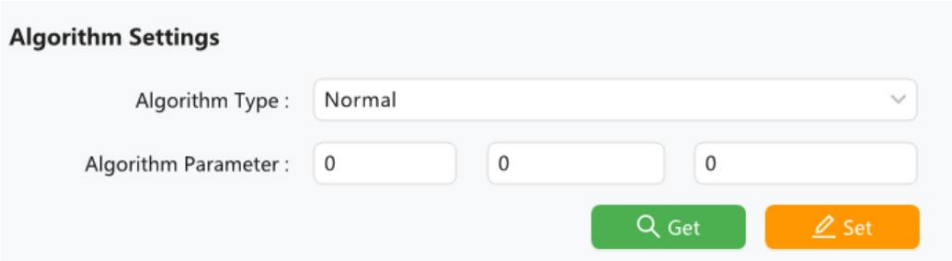
Click the "Get" button

to query the UHF Protocol Settings parameters. Click the "Set" button

to modify the device's UHF Protocol Settings parameters.

6.4.3.3. Algorithm Settings

The Algorithm Settings interface is shown below:



Algorithm Settings parameters include the following:

- Algorithm Type: The algorithm type value refers to the type of algorithm used by the device during inventory. The device may provide different inventory algorithms for different scenarios. Please refer to the device specification sheet for which algorithms the device supports.
- Algorithm Parameters: Algorithm parameters are configuration parameters for algorithm execution. Includes 3 parameters, with value ranges of 0~255, 0~65535, and 0~4294967295 respectively.

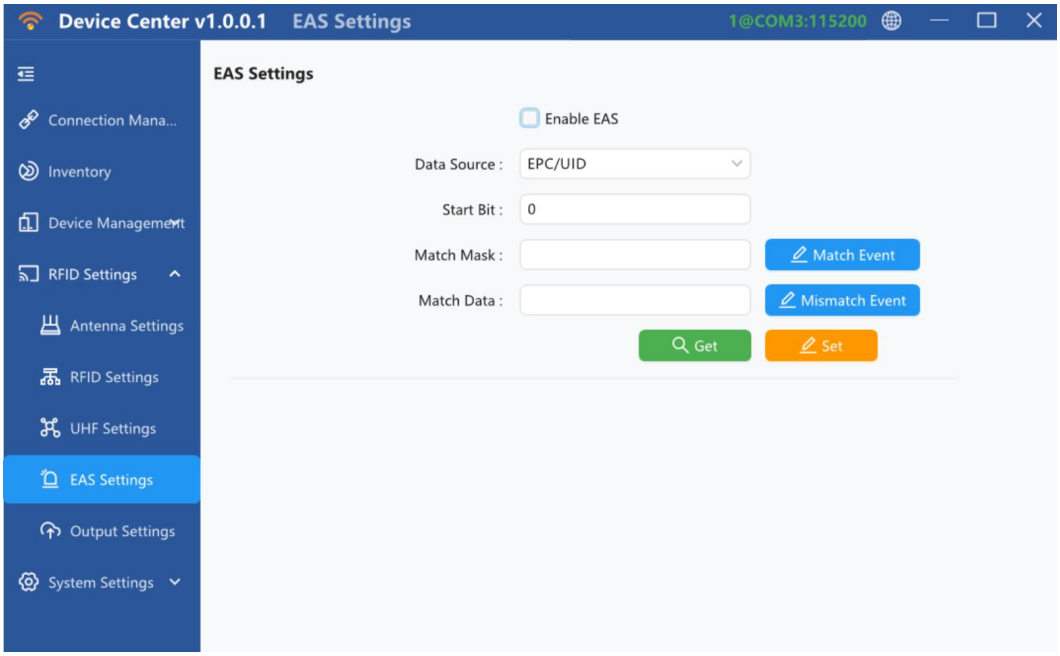
Click the "Get" button

to query the Algorithm Settings parameters. Click the "Set" button

to modify the device's Algorithm Settings parameters .

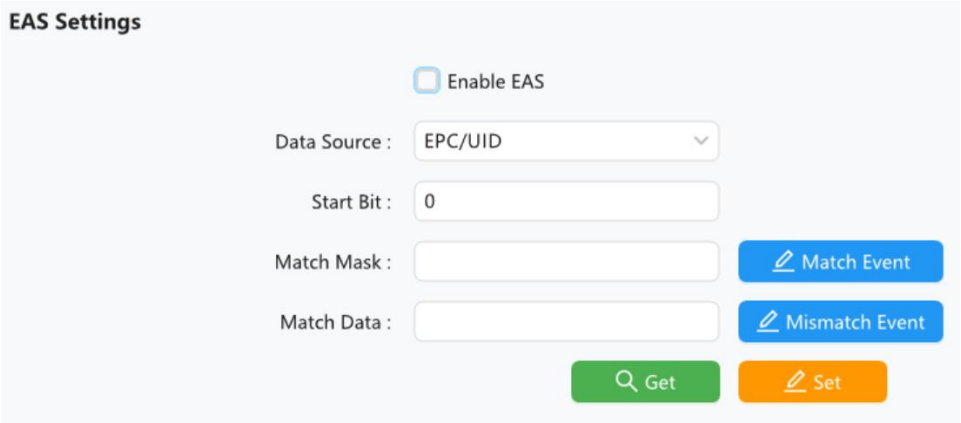
6.4.4. EAS Settings

The EAS Settings page covers EAS configuration. The EAS Settings page is shown below:



6.4.4.1. EAS Settings

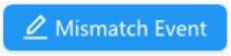
The EAS Settings interface is shown below:



EAS Settings parameters include the following:

- Enable EAS: Whether to enable the EAS function. After enabling EAS, each time a tag is read, the data corresponding to the data source and start bit will be ANDed with the match mask, and the result will be compared with the match data. If the comparison result is the same, a match event is triggered; otherwise, a non-match event is triggered.
- Data Source: The tag data source used by the EAS function. Can be EPC/UID, TID.

- Start Bit Address: The start bit address of the tag data used by the EAS function.
- Match Mask: The match mask used by the EAS function.
- Match Data: The match data used by the EAS function.
- Match Event: The event triggered by EAS when a match occurs. Click the "Match Event" button

to configure it.
- Non-Match Event: The event triggered by EAS when no match occurs. Click the "Non-Match Event" button 

to configure it.

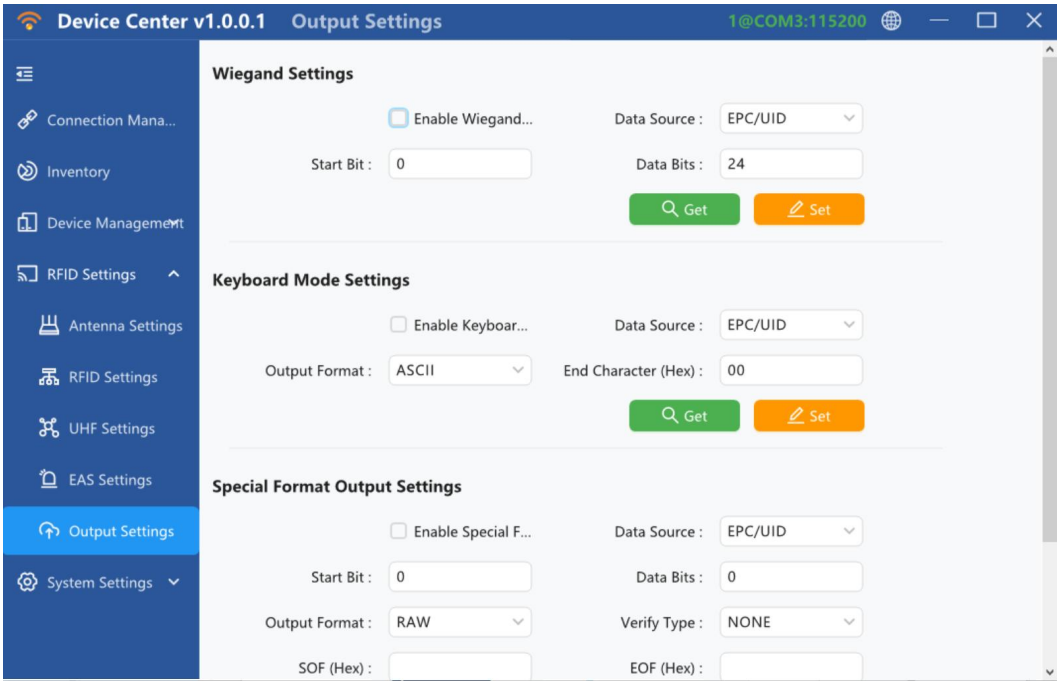
Click the "Get" button

to query the EAS Settings parameters. Click the "Set" button

to modify the device's EAS Settings parameters.

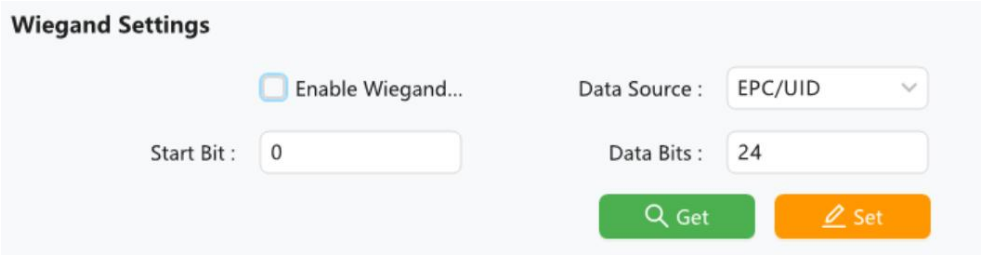
6.4.5. Output Settings

The Output Settings page covers functions such as Wiegand Settings, Keyboard Mode Settings, and Special Format Output Configuration. The Output Settings page is shown below:



6.4.5.1. Wiegand Settings

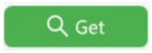
The Wiegand Settings interface is shown below:



Wiegand Settings parameters include the following:

- Enable Wiegand Output: Whether to enable the Wiegand output function. After enabling, each time a tag is read, the data corresponding to the data source and start bit will be sent out through the Wiegand port.
- Data Source: The tag data source used by the Wiegand function. Can be EPC/UID, TID.
- Start Bit Address: The start bit address of the tag data used by the Wiegand function. Set to -1 to send data from the end of the tag.

- Data Bit Length: The data bit length sent by Wiegand, supports 24, 32 bits, corresponding to Wiegand 26, 34.

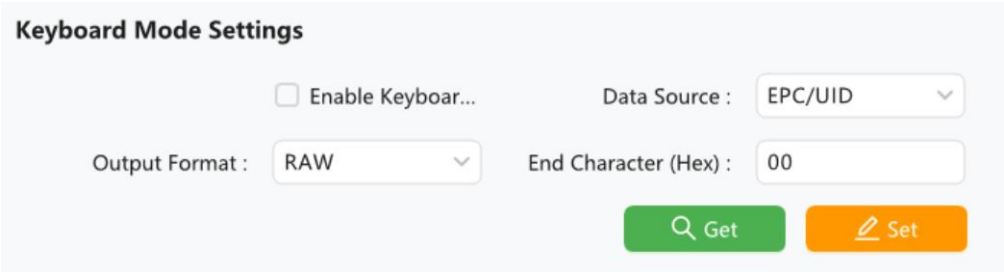
Click the "Get" button 

to query the Wiegand Settings parameters. Click the "Set" botton

to modify the device's Wiegand Settings parameters .

6.4.5.2. Keyboard Mode Settings

The Keyboard Mode Settings interface is shown below:



Keyboard Mode Settings parameters include the following:

- Enable Keyboard Output: Whether to enable the keyboard output function. After enabling, each time a tag is read, the data corresponding to the data source will be sent via the USB port in a simulated keyboard manner, according to the output format.
- Data Source: The tag data source used by the keyboard output function. Can be EPC/UID, TID.
- Output Format: The format conversion applied to the tag data when sent by the keyboard output function. Can be: RAW (raw data), ASCII (convert to ASCII characters).
- End Character: Additional data (in HEX format) sent after the keyboard output function finishes sending the tag data, such as 0A (Carriage Return), 09 (Tab), 20 (Space). If empty, it means no end character.

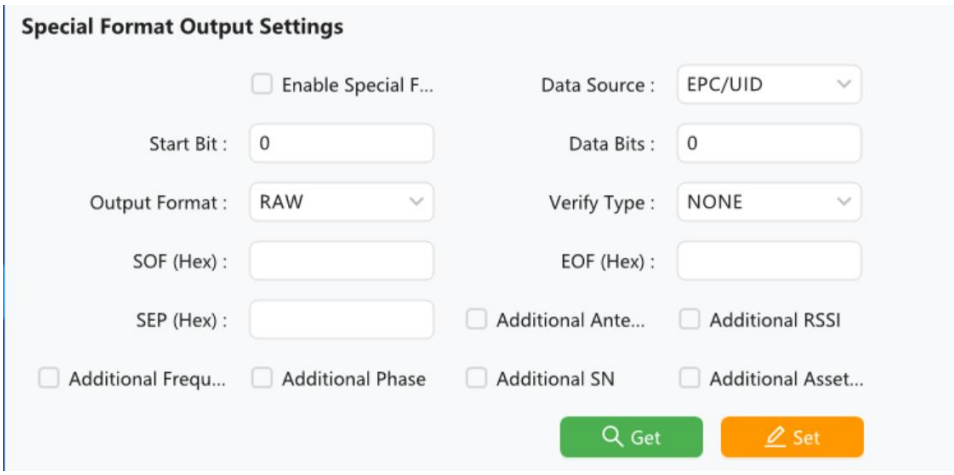
Click the "Get" button

to query the Keyboard Mode Settings parameters. Click the "Set" button

to modify the device's Keyboard Mode Settings parameters.

6.4.5.3. Special Format Output Settings

The Special Format Output Settings interface is shown below:



Special Format Output Settings parameters include the following:

- Enable Special Format Output: Whether to enable the special format output function. After enabling, each time a tag is read, the data corresponding to the data source will be output to the currently active communication connection in the specified output format (tag data is no longer output in the protocol format).
- Data Source: The tag data source used by the special format output function. Can be EPC/UID, TID.
- Start Bit Address: The start bit address of the tag data used by the special format output function.
- Data Bit Length: The data bit length of the tag data used by the special format output function.
- Output Format: The format conversion applied to the tag data during special format output. Can be: RAW (raw data), ASCII (convert to ASCII characters).

- Checksum Method: The frame checksum method used for special format output.
- Frame Header: The frame header used for special format output (HEX format).
- Frame Footer: The frame footer used for special format output (HEX format).
- Separator: The additional data separator used for special format output (HEX format).
- Additional Data: Additional data sent during special format output, including antenna number, RSSI, frequency, phase, serial number, asset number, etc.

Click the "Get" button

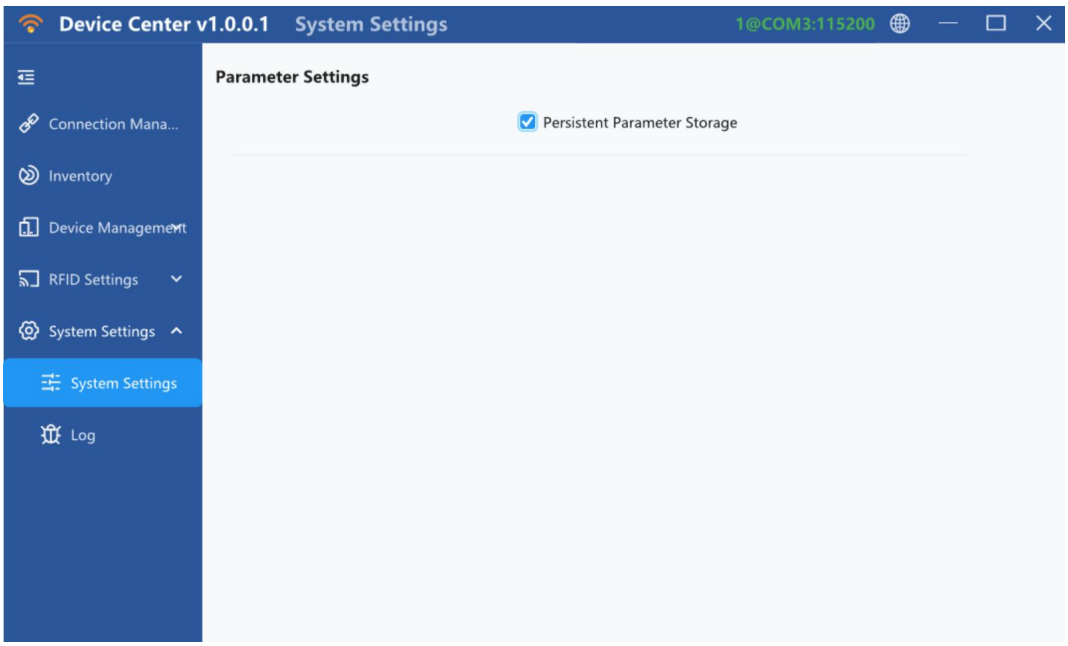
to query the Special Format Output Settings parameters. Click the "Set" button

to modify the device's Special Format Output Settings parameters.

6.5. System Settings

6.5.1. System Settings

System Settings cover parameter configuration. The System Settings interface is shown below:



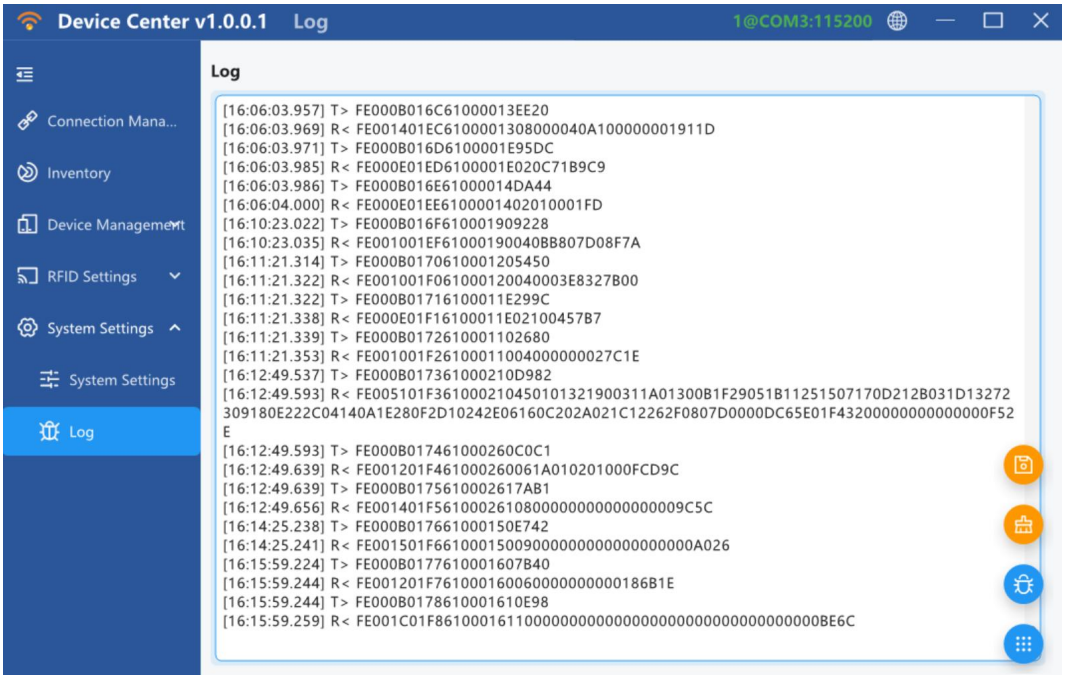
6.5.1.1. Parameter Settings

The Parameter Settings interface is shown below:

Parameter Settings include the parameter for persistent storage. It indicates whether to use persistent storage when setting device parameters. When checked, all parameter settings will be saved to the device's Flash and remain effective after power loss; otherwise, they will not be saved to Flash and will be lost after power loss.

6.5.2. Log

The Log page is used to display the communication log between the software and the device. The Log interface is shown below:



The Log page is also equipped with a set of floating action buttons. Users can click the button to expand or collapse them. These buttons cover functions such as Save, Clear, and other operations.

7. Common Issues & Solutions

7.1. Unable to Connect to Device via Serial Port

Please check if the physical connection between the device and the PC is reliable, if the serial port cable driver is correctly installed and recognized, and if the device address is set correctly (if the device address is unknown, try connecting using the broadcast address in a single-device scenario).

7.2. Unable to Connect to Device via Network Port

Please check if the device and the PC are on the same network, if the host computer can successfully ping the device, if the device is in TCP Client mode, and if the device address is set correctly (if the device address is unknown, try connecting using the broadcast address).

7.3. Unable to Find Device When Searching Network Port

Please check if the device and the PC are on the same network, and if the software is being blocked by the PC's firewall.

8. Security and Privacy

This software does not collect user privacy data; all operations are completed locally.